

Corporate Actions and Market Reactions: Evidence from Dividend Announcements on S&P 500 Returns.



Master's Thesis By

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PREFACE

This thesis is submitted in partial fulfilment of the requirements for a Master of Business Administration degree in Economics and Finance at Aalborg University, Denmark. The study was conducted during the academic year 2025/2026 under the supervision of Douglas Eduardo Turatti, to whom I owe a great deal of gratitude for his guidance, constructive feedback, and patience throughout the process.

This work represents one of the most demanding and rewarding intellectual challenges I have undertaken so far. The entire process taught me as much about persistence and problem solving as it did about financial economics.

I would like to express my sincere thanks to my family, particularly my Dad, big sister, my two brothers Ken, Theo and equally to my partner Chizo, whose encouragement and support from a distance meant more than words can say. I would also like to thank my colleagues and fellow students at Aalborg University for the many discussions and shared experiences that made this journey more enjoyable. Most importantly to my lovely mother Grace to whom I dedicate this work.

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Abstract

This thesis evaluates how the stock market reacts to Corporate actions (dividend announcements) made by firms in the S&P 500 over the period from January 2011 to December 2025. Using a sample of 22,118 ordinary dividend announcements across 414 firms, the study applies an event study methodology based on the market model to measure abnormal stock returns around each announcement date. Three types of dividend announcements are examined separately, dividend increases, dividend decreases and dividend initiations.

The results show that dividend increases generate a significant positive cumulative abnormal return of +0.27% over the (-3, +3) window (significant at the 1% level), with a weaker, marginally significant reaction of +0.13% over the short (-1, +1) window. For Increase, the market reaction is concentrated almost entirely on the day after the announcement, consistent with semi-strong form market efficiency. Dividend decreases produce a negative directional result but, the result does not reach statistical significance due to the small subsample of 244 events and high variation in individual reactions. Dividend initiations show an interesting intra-window pattern of pre-announcement gains followed by a sharp reversal on the announcement day, with no statistically significant cumulative result in either window.

A cross-sectional regression analysis finds that firm characteristics including size, profitability and valuation cannot explain variation in individual event returns, with R-squared values below 0.2% across all model specifications. This finding suggests that the magnitude of market reactions to individual announcements is driven primarily by event-specific and idiosyncratic factors that are not captured by standard accounting data.

The study contributes updated evidence that dividend announcements continue to carry meaningful informational content among large-cap U.S. firms, supporting the continued relevance of dividend signaling theory in a modern and information-rich market setting.

DISCLAIMER

In writing this thesis, I utilized R, a statistical software, for all data processing, event study analysis and cross-sectional regression modelling. Parts of the R code were developed with the assistance of Claude, an AI language model developed by Anthropic. All analytical results, interpretations and conclusions presented in this thesis are entirely the author's own.

Contents

CHAPTER 1.....	7
INTRODUCTION.....	7
1.1 BACKGROUND OF STUDY.....	8
1.2 PROBLEM STATEMENT.....	9
1.3 RESEARCH QUESTION.....	10
1.4 HYPOTHESIS.....	10
CHAPTER 2.....	11
LITERATURE REVIEW.....	11
2.1 CONCEPTUAL LITERATURE.....	12
2.1.1 DIVIDEND.....	12
2.1.2 DIVIDEND ANNOUNCEMENTS.....	12
2.1.3 DIVIDEND POLICY.....	13
2.1.4 DIVIDEND TIMELINE AND KEY DATES.....	13
2.1.5 LIFE-CYCLE OF DIVIDENDS.....	14
2.1.6 EFFICIENT MARKET HYPOTHESIS.....	14
2.1.7 SEMI-STRONG FORM.....	15
2.2 THEORETICAL LITERATURE.....	15
2.2.1 THE DIVIDEND SIGNALING THEORY.....	15
2.2.2 THE DIVIDEND IRRELEVANCE THEORY.....	16
2.2.3 THE AGENCY THEORY.....	16
2.3 EMPIRICAL LITERATURE.....	17
2.4 SYNTHESIS OF LITERATURE AND RESEARCH GAP.....	18
CHAPTER 3.....	18
METHODOLOGY.....	18
3.1 RESEARCH DESIGN.....	18
3.2 SAMPLE SELECTION.....	19
3.3 DATA SOURCES.....	21
3.4 EVENT STUDY METHODOLOGY.....	22
3.4.1 <i>Stock Returns</i>	22
3.4.2 <i>Event date (Announcement Date)</i>	23
3.4.3 <i>Estimation Window</i>	23
3.4.4 <i>Event Window</i>	23
3.4.5 <i>Expected Return Model</i>	24
3.4.6 <i>Abnormal Return</i>	24
3.4.7 <i>Cumulative Abnormal Return</i>	25

3.4.8 Average Abnormal Return (AAR).....	25
3.4.9 Cumulative Average Abnormal Return (CAAR).....	26
3.4.10 Event Study Variables.....	26
3.4.11 STATISTICAL TESTS.....	26
3.5 CROSS SECTIONAL ANALYSIS.....	27
3.5.1 Cross Sectional Regression Analysis.....	27
3.5.2 DEPENDENT VARIABLE.....	28
3.5.3 INDEPENDENT VARIABLES.....	28
3.5.4 Control Variables.....	29
Expected Signs of Coefficients.....	30
3.6 STATISTICAL TESTS.....	31
3.6.1 Test of Average Abnormal Returns (AAR).....	31
3.6.2 TEST OF CUMULATIVE AVERAGE ABNORMAL RETURNS (CAAR).....	32
3.6.3 SIGNIFICANCE LEVELS.....	32
3.6.4 REGRESSION STATISTICAL TESTS.....	32
3.7 SUMMARY OF METHODOLOGY.....	33
CHAPTER 4.....	34
EMPIRICAL RESULTS.....	34
4.1 SAMPLE DESCRIPTION AND DESCRIPTIVE STATISTICS.....	34
4.1.1 Descriptive Statistics of Cumulative Abnormal Returns.....	37
4.2 EVENT STUDY RESULTS.....	38
4.2.1 Average Abnormal Returns and Cumulative Average Abnormal Returns.....	38
4.2.2 HYPOTHESIS TESTS ON MEAN CUMULATIVE ABNORMAL RETURNS.....	42
4.2.3 Summary of Event Study Findings.....	45
4.3 DIAGNOSTIC TESTS.....	45
4.3.1 Normality of Cumulative Abnormal Returns.....	45
4.3.2 Multicollinearity Among Regression Variables.....	46
4.3.3 Pairwise Correlation of Continuous Control Variables.....	48
4.4 CROSS-SECTIONAL REGRESSION RESULTS.....	48
4.4.1 Panel A Event Window (-1, +1).....	50
4.4.2 Panel B Event Window (-3, +3).....	52
4.4.3 Industry Fixed Effects.....	53
4.4.4 Regression Diagnostics.....	54
4.4.5 Summary of Regression Findings.....	57
4.5 CHAPTER SUMMARY.....	58
CHAPTER 5.....	59
DISCUSSION.....	59
5.1 MARKET REACTION TO DIVIDEND INCREASE.....	60
5.2 MARKET REACTION TO DIVIDEND DECREASES.....	61
5.3 MARKET REACTION TO DIVIDEND INITIATIONS.....	63
5.4 CROSS-SECTIONAL REGRESSION RESULTS.....	64

5.5 IMPLICATIONS FOR MARKET EFFICIENCY	65
5.6 COMPARISON WITH PRIOR LITERATURE	66
5.7 CHAPTER SUMMARY	67
CHAPTER 6.....	68
LIMITATIONS.....	68
6.1 SAMPLE SCOPE AND DATA CONSTRAINTS	68
6.2 METHODOLOGY	69
6.3 STATISTICAL POWER.....	69
6.4 ABSENCE OF AN EXPECTATIONS MODEL	69
CHAPTER 7.....	70
CONCLUSION	70
7.1 SUMMARY OF KEY FINDINGS	70
7.2 THEORETICAL CONTRIBUTIONS.....	71
7.3 PRACTICAL IMPLICATIONS	72
7.4 RECOMMENDATIONS	72
7.5 CLOSING STATEMENT	73
REFERENCES	74
APPENDIX	77

CHAPTER 1

Introduction

Corporate actions are an important channel through which firms communicate information to the market. Through these decisions which can include earnings announcements, dividend declarations, share repurchases, mergers and acquisitions, and equity issuances firms disclose information that may alter their financial structure, cash-flow distribution or strategic direction. Because such actions often involve forward-looking managerial judgement, they are closely monitored by investors and analysts as potential sources of value-relevant information.

In efficient capital markets, stock prices are expected to adjust to the extent that these events convey new or unexpected information about firm fundamentals. Among all corporate actions, dividend announcements occupy a particularly central role because they directly affect shareholders' cash flows and may reflect management's private assessment of the firm's current performance and future prospects.

Classical financial theory suggests that under perfect market conditions, dividend policy should be irrelevant for firm value and announcements should not generate abnormal returns (Miller and Modigliani, 1961). However, extensive empirical evidence has challenged this prediction. When information between managers and investors is unequal, dividend changes can serve as credible signals of a firm's future cash-flow prospects. Because sustaining higher dividend payments is costly, only financially strong firms can credibly commit to them, meaning dividend increases should be associated with positive abnormal returns and decreases with negative reactions (Bhattacharya, 1979; Miller and Rock, 1985). Dividend policy may also serve an agency function by reducing the cash available for managers to spend on projects that do not benefit shareholders, making dividend increases and initiations a signal of managerial discipline (Jensen, 1986).

This study distinguishes between three types of dividend announcements, increases, decreases and initiations because each carries a different signal to the market. Increases suggest confidence in future earnings, decreases may indicate financial pressure, and initiations often signal that a firm has reached a level of financial maturity where it can commit to regular shareholder payments (Grullon, Michaely and Swaminathan, 2002). Treating all announcements as a single category would obscure these differences and produce results that are difficult to interpret.

The current study examines how equity markets respond to these three types of dividend announcements by S&P 500 firms over the period January 2011 to December 2025, using an event study framework. S&P 500 firms are large and extensively covered by analysts, making this an especially demanding test of whether dividend announcements still carry informational content in a highly efficient market setting.

The contribution of this study is threefold. First, it provides updated evidence on dividend announcement effects using recent data for S&P 500 firms, a re-examination that is warranted given how payout policies and investor behaviour have evolved alongside the growing use of share repurchases. Second, it distinguishes explicitly between increases, decreases and initiations, allowing for a more detailed analysis of how markets interpret different dividend signals. Third, by focusing on a highly liquid and information-rich market, it contributes to the debate on market efficiency by testing whether abnormal returns persist even for the most widely followed firms in the world.

1.1 Background of Study

Dividends have historically represented one of the primary mechanisms through which corporations distribute profits to shareholders. Long before the development of modern capital markets, dividend payments served as a tangible return on investment, particularly for investors seeking regular income. Although corporate payout practices have evolved over time, most notably with the increasing use of share repurchases, dividends remain a central component of payout policy for many publicly listed firms. This is especially true for large and mature companies, for which dividend payments continue to signal financial stability, sustainable earnings and long-term profitability.

In financial markets, dividend announcements constitute discrete and well-defined corporate events that are closely followed by investors and financial analysts. Such announcements convey information not only about the level of current payouts but also about management's expectations regarding future profitability and cash-flow sustainability. Consequently, dividend announcements have the potential to influence investor perceptions and trigger adjustments in stock prices. Understanding how markets respond to these announcements is therefore important for evaluating the informational role of corporate payout decisions.

The relevance of dividend announcements has persisted even as capital markets have become increasingly efficient and information dissemination has accelerated. Advances in technology, greater analyst coverage and improved disclosure standards have reduced information asymmetries, particularly in developed markets such as the United States. Nevertheless, empirical research continues to document market reactions to corporate announcements, raising questions about whether dividend-related information is fully anticipated or whether it continues to convey incremental insights to market participants.

Firms included in the S&P 500 index provide a particularly informative setting for examining these issues. These firms are typically large, widely held and subject to extensive scrutiny by institutional investors and financial analysts. Moreover, they account for a substantial share of total market capitalization and dividend payments in the U.S. equity market. Dividend announcements by these firms therefore have the potential to influence not only individual stock prices but also broader market sentiment. Studying dividend announcements in this context allows for an assessment of whether market reactions persist even among firms operating in a highly transparent and information-rich environment.

Against this background, analyzing stock market reactions to dividend announcements offers valuable insights into the functioning of modern capital markets. Examining abnormal stock returns around dividend announcement dates helps to clarify whether dividend policy decisions convey information beyond what is already reflected in prices. Such analysis contributes to a deeper understanding of investor behavior, corporate communication and the extent to which financial markets efficiently process publicly available information.

1.2 Problem Statement

Despite the long-standing importance of dividend policy in corporate finance, there remains no clear consensus on how financial markets interpret dividend announcements. Classical financial theory suggests that dividend announcements should not affect firm value, while alternative perspectives emphasize their informational and disciplinary roles. This theoretical divergence raises an important empirical question regarding whether dividend announcements continue to influence stock prices in modern financial markets.

The empirical evidence on this question remains mixed. While a number of studies report significant abnormal stock returns around dividend announcement dates (Aharony and Swary, 1980; Michaely, Thaler and Womack, 1995), other research suggests that the information content of dividend announcements has declined over time as markets have become more efficient and institutional investor participation has grown (Amihud and Li, 2006). This raises the question of whether dividend-related information is already anticipated by investors and incorporated into stock prices prior to the formal announcement.

A further limitation of the existing literature is that much of it treats different types of dividend announcements in isolation. Studies have analyzed the stock price effects of dividend increases and decreases around announcement dates (Lonie et al., 1996), while other research has focused specifically on dividend initiations and omissions (Michaely, Thaler and Womack, 1995). Dividend increases, decreases and initiations may convey fundamentally different signals regarding a firm's financial condition and future prospects, yet their effects are rarely examined within a single unified empirical framework.

These gaps are particularly relevant in the context of large, highly followed firms such as those in the S&P 500 index. If dividend announcements generate abnormal returns even in such a transparent and information-rich environment, it would suggest that dividend policy continues to convey value-relevant information beyond what is already reflected in prices. Furthermore, this study addresses the question of whether dividend announcements made by S&P 500 firms generate abnormal stock returns and whether these reactions differ systematically across dividend increases, decreases, and initiations. By doing so, the study contributes to the literature by providing updated evidence using recent data and by explicitly differentiating between announcement types within a single empirical framework.

1.3 Research Question

Building on the theoretical and empirical background outlined above, this study addresses the following primary research question:

"Do dividend announcements by S&P 500 firms generate abnormal stock returns, and does the magnitude and direction of these returns differ systematically across dividend increases, decreases and initiations?"

To address this question, the following supporting questions guide the empirical analysis:

1. Does a dividend increase announcement generate positive abnormal stock returns around the announcement date?
2. Does a dividend decrease announcement generate negative abnormal stock returns around the announcement date?
3. Does a dividend initiation announcement generate significant abnormal stock returns around the announcement date?

1.4 Hypothesis

Based on the theoretical frameworks discussed earlier in 1.1 and 1.2, dividend announcements may influence stock prices if they convey new or unexpected information to investors. Under the dividend irrelevance proposition of Franco Modigliani and Merton Miller (1961), dividend policy should not affect firm value in perfect capital markets. However, when market imperfections such as information asymmetry and agency conflicts are present, dividend changes may convey information about a firm's future prospects (Bhattacharya, 1979). Empirical studies have indeed documented significant stock price reactions around dividend announcements, suggesting that such events can carry informational content for investors (Aharony & Swary, 1980; Michaely, Thaler & Womack, 1995).

Dividend increases are commonly interpreted as favorable signals regarding a firm's financial strength and future cash-flow sustainability. According to dividend signaling theory, managers are unlikely to increase dividends unless they are confident that the firm can sustain higher payouts in the future (Bhattacharya, 1979). Empirical evidence supports this interpretation, as dividend increases are frequently associated with positive abnormal stock returns around announcement dates (Aharony & Swary, 1980; Grullon, Michaely & Swaminathan, 2002). Accordingly, the first hypothesis is formulated as follows:

H1: Dividend increase announcements by firms in the S&P 500 index are associated with positive abnormal stock returns around the announcement date.

In contrast, dividend decreases or omissions are generally perceived as unfavorable events. A reduction in dividend payments may signal deteriorating profitability, financial constraints, or lower expected future earnings. From both signaling and agency perspectives, such announcements raise concerns about a firm's financial condition and managerial expectations regarding future performance, and empirical studies confirm that dividend cuts typically generate negative abnormal returns reflecting unfavorable revisions in investor expectations (Healy and Palepu, 1988; Michaely, Thaler and Womack, 1995) This leads to the second hypothesis:

H2: Dividend decrease announcements by firms in the S&P 500 index are associated with negative abnormal stock returns around the announcement date.

Dividend initiations represent a distinct category of dividend announcements because they introduce dividend payments where none previously existed. Initiating a dividend may signal that a firm has reached a stage of financial maturity characterized by stable earnings and stronger cash-flow generation (Grullon et al., 2002). Empirical research shows that dividend initiations are generally associated with positive stock price reactions, suggesting that investors interpret them as favorable signals about long-term financial prospects (Michaely et al., 1995). This motivates the third hypothesis:

H3: Dividend initiation announcements by firms in the S&P 500 index generate positive abnormal stock returns around the announcement date.

Finally, given the informational richness and high level of analyst coverage associated with S&P 500 firms, any observed abnormal returns around dividend announcement dates would suggest that dividend-related information is not fully anticipated by the market. Differences in market reactions across dividend increases, decreases, and initiations would further indicate that investors distinguish between types of dividend announcements when forming expectations. This consideration motivates a comparative assessment across announcement types rather than a uniform treatment of dividend events.

CHAPTER 2

Literature Review

This chapter reviews the theoretical and empirical foundations relevant to stock price reactions to dividend-related information. It is organized into three sections. The first section presents some conceptual definitions, the second section reviews the main theoretical frameworks explaining how dividend announcements may influence firm value and stock prices, particularly dividend irrelevance, signaling, and agency theories. The third section examines empirical evidence on stock market reactions to dividend announcements. Together,

these discussions establish the foundation for analyzing dividend announcement effects among firms included in the S&P 500 index.

2.1 Conceptual Literature

In this section, we look at some of the key concepts that underpin the analysis of dividend announcements and stock market reactions. Establishing precise definitions is essential for distinguishing between closely related corporate payout mechanisms and for framing the empirical analysis within an event study context.

2.1.1 Dividend

A dividend is a distribution of a firm's earnings or accumulated reserves to its shareholders, typically in the form of cash payments. Dividends represent a direct return on equity investment and are one of the primary corporate payout mechanisms. In the U.S. equity market, dividends are commonly paid on a quarterly basis and must be formally declared by a firm's board of directors, with disclosure requirements governed by securities regulations enforced by the Securities and Exchange Commission (SEC).

The determinants of dividend decisions were extensively examined by Lintner (1956), who shows that firms adjust dividends cautiously and are generally reluctant to reduce them because dividend cuts are often interpreted by investors as negative signals regarding future performance. Consistent with this observation, Brav et al. (2005) report that managers place significant emphasis on maintaining dividend stability due to concerns about market reactions. Building on this perspective, Denis and Osobov (2008) document that dividend-paying firms are typically more mature and characterized by stable cash flows, while DeAngelo, DeAngelo, and Stulz (2006) argue that the likelihood of paying dividends increases as firms progress through their life cycle and accumulate retained earnings that exceed their investment needs.

2.1.2 Dividend Announcements

Dividend announcements are corporate disclosures through which firms communicate dividend decisions to the market. These announcements provide new or updated information about dividend payments and may therefore influence investors' expectations regarding a firm's future cash flows and financial prospects. As Healy and Palepu (1988) explain, dividend announcements may take several forms, including dividend initiations, increases, decreases, and omissions, each potentially conveying different information to investors.

From a market efficiency perspective, dividend announcements represent clearly identifiable information events that allow researchers to examine how quickly stock prices incorporate new public information. According to Fama (1970), under the semi-strong form of the Efficient Market Hypothesis, stock prices should adjust rapidly to publicly available information. Consequently, abnormal returns associated with dividend announcements are expected to occur around the announcement date.

For this reason, dividend announcements are commonly examined using event study methodology, which evaluates stock price movements within short windows surrounding the announcement date. As discussed by MacKinlay (1997), this approach allows researchers to isolate market reactions to specific information events. Empirical evidence documented by Aharony and Swary (1980) further shows that dividend increases and initiations tend to generate positive abnormal returns, while dividend decreases or omissions are often associated with negative price reactions.

2.1.3 Dividend Policy

Dividend policy refers to a firm's long-term strategy regarding the distribution of earnings to shareholders. It involves managerial decisions about whether dividends should be paid, the level and frequency of payouts, and how payments are adjusted over time. As discussed by Brealey, Myers, and Allen (2020), dividend policy reflects management's evaluation of profitability, expected future cash flows, liquidity conditions, investment opportunities, and shareholder preferences.

An important characteristic of dividend policy is its emphasis on stability. In his seminal study, Lintner (1956) finds that firms adjust dividends gradually and avoid frequent reductions because dividend cuts may negatively affect investor confidence.

Dividend policy is also closely linked to a firm's investment and financing decisions. Firms with limited investment opportunities may distribute excess cash to shareholders and vice versa. Understanding dividend policy is therefore essential for interpreting dividend announcements and their implications for stock market behavior.

2.1.4 Dividend timeline and Key dates

Dividend payments follow a structured timeline that determines both the spread of information and shareholder eligibility.

The process begins with the announcement date, when the firm's board of directors formally declares the dividend and specifies the dividend amount, ex-dividend date, record date, and payment date.

The ex-dividend date is the first trading day on which the stock is sold without the right to receive the declared dividend. Investors who purchase the stock on or after this date are not entitled to the dividend, while those who hold the stock before the ex-dividend date retain the claim.

The record date identifies the shareholders eligible to receive the dividend, and the payment date is when the dividend is distributed. Because these events are largely anticipated, they typically generate limited market reactions compared to the announcement date.

2.1.5 Life-Cycle of Dividends

The life-cycle theory of dividends explains how firms' payout policies evolve as they progress through different stages of development. Rather than treating dividend policy as static, the theory suggests that firms' decisions to distribute earnings depend on their maturity, profitability, and investment opportunities, as discussed by Berk and DeMarzo (2023).

In the early stages of the corporate life cycle, firms typically face high growth opportunities, significant financing needs, and relatively volatile cash flows. As Fama and French (2001) observe, young firms are therefore less likely to pay dividends because internal funds are prioritized for reinvestment and expansion. As firms mature, however, profitability and internal cash generation tend to become more stable while investment opportunities decline. Under such conditions, distributing excess cash through dividends becomes a natural component of financial policy.

Empirical research supports this perspective. DeAngelo, DeAngelo, and Stulz (2006) show that dividend payments are more common among mature firms with substantial accumulated earnings, while Grullon, Michaely, and Swaminathan (2002) link dividend increases to firms experiencing declining investment intensity and greater cash-flow stability.

An important implication of the life-cycle theory is that dividend initiations may carry a distinct economic meaning compared to routine dividend increases. According to Fama and French (2001), firms often initiate dividends when they transition from a growth-oriented phase to a more mature stage characterized by stable profitability and fewer investment opportunities. As a result, dividend initiations may signal not only current earnings strength but also a structural shift in the firm's financial maturity, which may lead investors to react differently to initiation announcements compared with routine dividend changes.

2.1.6 Efficient Market Hypothesis

The Efficient Market Hypothesis (EMH) explains how information is reflected in security prices. According to Fama (1970), an efficient market is one in which asset prices fully incorporate available information, making it impossible for investors to consistently earn abnormal returns using publicly known data. When new information becomes available, prices should adjust rapidly to reflect its implications for firm value.

EMH is typically classified into three forms based on the scope of information reflected in prices. Weak-form efficiency suggests that prices incorporate all historical market data, such as past prices and trading volumes. Semi-strong-form efficiency proposes that prices reflect all publicly available information, including corporate announcements such as dividends and earnings. Strong-form efficiency extends this view by suggesting that prices reflect both public and private information, implying that even insider information cannot generate consistent abnormal returns, as discussed by Malkiel (1989).

2.1.7 Semi-Strong Form

The semi-strong form of the Efficient Market Hypothesis is particularly relevant for analyzing stock price reactions to corporate announcements. As originally proposed by Fama (1970), semi-strong efficiency implies that security prices adjust rapidly to publicly available information. Consequently, investors cannot systematically earn abnormal returns using information disclosed through corporate announcements such as earnings releases, mergers, or dividend declarations.

Dividend announcements provide a useful setting for examining this principle because they represent clearly timed public disclosures that may alter investors' expectations regarding a firm's future cash flows. If a dividend announcement contains information that differs from market expectations, stock prices should adjust immediately around the announcement date. Conversely, if the announcement merely confirms existing expectations, no abnormal price reaction should occur because the information would already be incorporated into prices.

This framework is particularly relevant for firms included in the S&P 500 index, which are characterized by high trading liquidity, extensive analyst coverage, and frequent information disclosure. As MacKinlay (1997) notes, such environments provide an appropriate setting for event study analysis, allowing researchers to evaluate how quickly and accurately markets incorporate new public information.

2.2 Theoretical Literature

The theories in this section address the mechanisms through which dividend announcements may influence investor beliefs and asset prices. In particular, the literature draws on models of information asymmetry, market efficiency, agency considerations, and behavioral responses to explain why dividend announcements may generate abnormal stock returns, despite the mechanical nature of dividend payments. These theories provide a firm basis for the empirical analysis of this work and guide the interpretation of the results presented later in this study.

2.2.1 The Dividend Signaling Theory

Dividend signaling theory provides a prominent explanation for why dividend announcements may influence stock prices. The theory is based on information asymmetry between managers and investors, where managers possess superior knowledge about the firm's current performance and future prospects. As proposed by Bhattacharya (1979) and John and Williams (1985), managers may use dividend changes to signal private information about expected future cash flows.

Under this framework, investor reactions depend on whether the dividend announcement conveys information that was not previously anticipated. An unexpected dividend increase may lead investors to revise their

expectations of future profitability upward, while dividend reductions or omissions may signal weaker financial prospects. These belief revisions are reflected in stock price adjustments around the announcement date, generating abnormal returns when the announcement differs from market expectations, as explained by Miller and Rock (1985).

Importantly, signaling theory does not imply that all dividend announcements will affect stock prices. Consistent with semi-strong-form market efficiency, only announcements that reveal new information are expected to generate price reactions. Consequently, empirical studies often focus on the unexpected component of dividend announcements when analyzing market responses.

2.2.2 The Dividend Irrelevance Theory

Dividend irrelevance theory provides a benchmark for understanding when dividend announcements should not influence stock prices. Developed by Miller and Modigliani (1961), the theory argues that in perfect capital markets a firm's dividend policy does not affect its market value. Instead, firm value is determined solely by its investment decisions and the cash flows generated by its assets.

This proposition relies on several strong assumptions, including the absence of taxes and transaction costs, frictionless trading, and symmetric information between managers and investors. Under such conditions, investors can create their preferred income streams through trading, a mechanism often referred to as "homemade dividends," as discussed by Brealey, Myers, and Allen (2020).

From this perspective, dividend announcements should not generate abnormal stock price reactions unless they convey new information about firm fundamentals. Consistent with Fama's (1970) framework of market efficiency, only the unexpected component of an announcement should influence stock prices. Consequently, the presence of abnormal returns around dividend announcements would suggest that at least one of the perfect-market assumptions underlying the irrelevance theory does not hold in practice.

2.2.3 The Agency Theory

In publicly traded companies, shareholders delegate decision-making authority to managers, which creates the potential for conflicts of interest when managerial objectives diverge from shareholder value maximization. As explained by Jensen and Meckling (1976), these agency conflicts arise because managers may have incentives to retain and allocate corporate resources in ways that do not necessarily maximize shareholder wealth.

Dividend payments can serve as a mechanism to reduce such agency problems. When firms generate cash flows that exceed profitable investment opportunities, retaining these funds may increase the risk of inefficient spending or value-destroying investments. Jensen (1986) argues that distributing excess cash to shareholders limits managerial discretion over free cash flow and reduces the scope for agency costs.

From this perspective, dividend announcements may also act as governance signals. An unexpected dividend increase may indicate that management is committed to limiting internal cash accumulation and aligning decisions with shareholder interests, potentially leading to positive stock price reactions. Conversely, dividend reductions or omissions may raise concerns about increased internal cash retention and weaker managerial discipline.

Unlike signaling theory, which emphasizes information about future earnings, agency theory interprets dividend changes as mechanisms that influence expectations about managerial behavior and capital allocation. As noted by Allen and Michaely (2003), investors may therefore react to dividend announcements because they reflect changes in governance and the control of managerial discretion.

2.3 Empirical Literature

A large body of empirical research has examined stock market reactions to dividend announcements using event study methodologies. Early empirical studies provide evidence that dividend announcements may convey valuable information to investors. Aharony and Swary (1980) investigated market reactions to quarterly dividend and earnings announcements and found significant abnormal returns around dividend announcement dates. Their results suggested that dividend announcements contain information relevant for investors beyond earnings disclosures.

Subsequent research further explores the informational content of dividend changes. Healy and Palepu (1988) examine dividend initiations and omissions and report that firms initiating dividends tend to experience positive abnormal returns, while firms omitting dividends experience negative stock price reactions. Similarly, Michaely, Thaler, and Womack (1995) find strong positive market reactions to dividend initiations and negative reactions to dividend omissions, providing evidence that investors interpret these events as signals about future firm performance.

Empirical studies have also investigated the relationship between dividend changes and firm life-cycle characteristics. Grullon, Michaely, and Swaminathan (2002) show that dividend increases are often associated with firms transitioning toward a more mature stage of development characterized by declining investment opportunities and more stable cash flows. Their findings suggest that dividend increases may signal structural changes in firm maturity rather than merely short-term earnings expectations.

More recent empirical research has revisited the informational role of dividends in modern financial markets. Denis and Osobov (2008) analyze international evidence on dividend policy and find that firms with strong profitability and limited investment opportunities are more likely to distribute dividends. However, some studies argue that the informational content of dividend announcements may have weakened over time as financial markets have become more transparent and investors have access to a broader set of information sources (Benartzi, Michaely, and Thaler, 1997).

2.4 Synthesis of Literature and Research Gap

The literature reviewed in this chapter highlights dividend announcements as important corporate events that may influence investor expectations and stock prices. Conceptual discussions emphasize that dividend announcements represent publicly disclosed decisions, while the Efficient Market Hypothesis provides a benchmark for understanding how such information should be incorporated into prices. In particular, the semi-strong form of market efficiency suggests that stock price reactions to dividend announcements should occur rapidly and reflect the unexpected component of the information released.

Theoretical perspectives offer different explanations for these reactions. While the dividend irrelevance proposition predicts no market response under perfect capital market conditions, alternative theories based on information asymmetry, agency considerations, and firm life-cycle dynamics suggest that dividend changes may convey information about future cash flows, managerial discipline, or firm maturity. Consequently, whether dividend announcements generate abnormal returns remains an empirical question.

Existing empirical studies generally document stock price reactions around dividend announcements, although the magnitude and persistence of these effects vary across studies and time periods. Some evidence also suggests that the informational role of dividend announcements may have declined in more recent years, particularly in highly transparent markets.

Against this background, relatively limited attention has been devoted to systematically comparing market reactions to dividend increases, decreases, and initiations within a unified framework using recent data from highly information-rich environments such as the S&P 500 index. The present study addresses this gap by examining stock price reactions to different types of dividend announcements among S&P 500 firms using an event study methodology.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study adopts a quantitative empirical research design to test whether dividend announcements by S&P 500 firms generate abnormal stock returns, and whether these reactions differ systematically across dividend increases, decreases, and initiations. The design follows directly from the research question set out in Section 1.3 and operationalizes the hypotheses developed in Section 1.4.

The analysis proceeds in two stages. The first stage applies an event study to measure the market's reaction to each announcement, comparing realised returns around the announcement date with the returns that would have been expected in its absence (MacKinlay, 1997). This stage tests H1–H3 by establishing whether abnormal returns arise and in which direction. The second stage uses a cross-sectional regression, with the cumulative abnormal return as the dependent variable, to examine whether the three announcement types elicit significantly different reactions once firm characteristics, namely size, profitability, leverage, valuation, and industry, are held constant. The event study therefore identifies whether the market reacts, while the regression explains how that reaction varies across announcement types and firms.

Two design choices reflect the contribution outlined in Chapter 1. First, restricting the sample to S&P 500 firms provides a deliberately demanding test of informational content: because these firms are large, liquid, and heavily covered by analysts, any abnormal returns would indicate that dividend announcements convey information not already impounded in prices. Second, treating increases, decreases, and initiations as distinct categories rather than a single event class allows the differing signals identified in the literature to be tested directly within one framework. The detailed estimation procedure, the construction of abnormal and cumulative abnormal returns, and the specification of the estimation and event windows are set out in the event study methodology section that follows.

3.2 Sample Selection

An important component of empirical research is constructing a sample that adequately represents the population under study while minimizing potential selection bias.

The sample consists of dividend announcement events made by firms that were constituents of the S&P 500 index at the time of the announcement. Using the S&P 500 as the sampling frame is appropriate because the index comprises large, publicly traded U.S. firms characterized by substantial trading volume, broad analyst coverage, and high institutional ownership. These characteristics create an information-rich and relatively efficient market environment, making the S&P 500 an appropriate setting for testing whether dividend announcements convey incremental information to investors.

Due to the unavailability of historical S&P 500 membership data, the sample is constructed using a fixed list of index constituents as of 2 February 2025, the date on which the constituent data was extracted from FactSet. All firms included in the sample were verified S&P 500 members on that reference date, and their dividend announcement histories were then traced back to the start of the study period on 1 January 2011.

The sample period spans from 1 January 2011 to 31 December 2025. This 15-year period is sufficiently extensive to capture a large number of dividend announcement events across varying market conditions, thereby improving the statistical power and reliability of the event study analysis. Additionally, the selected period reflects a modern market environment characterized by rapid information dissemination, algorithmic trading,

and increased institutional investor participation. Using recent data also allows the study to contribute updated evidence to the dividend signaling literature, particularly in light of findings suggesting that the informational content of dividend announcements may have changed over time (Amihud & Li, 2006).

First, all dividend announcements made by S&P 500 constituent firms during the sample period are identified and extracted from FactSet. Second, each dividend announcement is classified into one of three categories: dividend increase, dividend decrease, or dividend initiation, based on changes relative to the firm's previous dividend announcement. This classification enables comparison of market reactions across different types of dividend events.

Several screening criteria are applied to improve the reliability of the dataset. First, dividend events are excluded on an event-by-event basis if insufficient stock price data are available to estimate the market model over the estimation window of $t-250$ to $t-30$. Specifically, a minimum of 60 daily return observations is required within the estimation window for a given event to be retained. This threshold ensures that the OLS estimates of alpha and beta are statistically reliable. Importantly, this criterion is applied at the individual event level rather than at the firm level, meaning a firm is not excluded from the sample entirely if some of its events lack sufficient data, only the specific events with inadequate observations are dropped. This approach maximizes the number of usable observations while maintaining estimation quality.

Secondly, observations are excluded if the relevant financial data required for the cross-sectional regression analysis specifically firm size, profitability, and price-to-book ratio are unavailable for the fiscal year preceding the announcement date. These observations are retained in the event study but excluded only from the regression analysis, ensuring that the event study results reflect the broadest possible sample while the regression sample maintains internal consistency.

After applying these criteria, the final event study sample consists of all ordinary dividend announcements by S&P 500 constituent firms between 1 January 2011 and 31 December 2025 for which the minimum estimation window data requirement is met. The regression subsample is further restricted to observations with complete control variable data. No additional filters are applied for confounding events, incomplete return histories, or firm-level data gaps, as the event-by-event approach naturally accommodates firms with partial data availability across the study period.

The use of a fixed constituent list as of 2 February 2025 introduces two potential limitations that warrant discussion.

First, survivorship bias may arise because the sample includes only firms that were S&P 500 members on the reference date, thereby excluding firms that were constituents at some point during the 2011–2025 study period but were subsequently removed due to bankruptcy, acquisition, delisting, or index reconstitution prior to February 2025. If such removed firms systematically differed in their dividend behaviour or market reactions

from those that survived, the estimated abnormal returns may overstate the market reaction relative to what would be observed using a fully dynamic constituent list.

Second, a mild form of look-ahead bias is present, as firms are selected based on their membership status at a date that falls within the study period rather than at its beginning. Consequently, the sample implicitly conditions on firms having remained in the index through February 2025, which was unknown at earlier points in the sample period.

However, both concerns are mitigated by the nature of the research question. The study focuses on the informational content of individual dividend announcement events specifically whether announcements generate abnormal returns rather than on index composition or cross-sectional performance over time.

Since the market model is estimated and abnormal returns are calculated on a firm-by-firm, event-by-event basis using only data available prior to each announcement, the core empirical estimates are unlikely to be materially distorted by the fixed constituent approach. This limitation is nonetheless acknowledged as a data constraint and should be considered when interpreting and generalizing the findings.

3.3 Data Sources

This study's analysis is done on secondary data gained through a variety of established finance databases with FactSet being the main source of data for this work.

The S&P 500 constituent list used in this study reflects index membership as of 2 February 2025, which serves as the fixed reference date for sample construction, as historical membership data was not available for the full study period.

In cases when there is a lack of access to FactSet, an approved alternative publicly accessible data sources like Yahoo Finance was utilized to either verify or supplement the collection of data within the study.

Information in this dataset has been combined to create a single set of data from different sources. Dividend announcement information was the first source obtained using the corporate actions module to find dividend declaration and dividend per share dates to use in receipt of identifying event dates in an event study analysis.

Daily adjusted closing prices and daily return data are used to obtain stock price information from Factset. These daily price data were used to calculate stock returns prior to estimating abnormal returns around dividend announcement event dates. The next source of information used for this study was market index data such as the S&P 500 index; therefore, returns measured using a market index will be used to determine expected returns using the market model, which will assist in identifying a firm's abnormal performance after controlling for overall market movements.

Daily adjusted closing prices are extracted for each firm over the period from 31 December 2009 to 31 December 2025. The starting point of 31 December 2009 is selected to ensure that sufficient price history is

available to construct the estimation window for dividend announcement events occurring at the beginning of the study period. Specifically, given that the estimation window spans from t-250 to t-30 trading days prior to each announcement date, firms with announcements in early 2011 require price data extending back to approximately late 2009 to satisfy the minimum estimation window requirement.

The ending point of 31 December 2025 corresponds to the final day of the study period. All daily returns are computed from these adjusted closing prices using the formula described in Section 3.4.1, with the S&P 500 index used as the market benchmark throughout.

To analyze how firm-level financial attributes relate with stock price responses to dividend announcement events, firm-specific financial attributes have been gathered from Factset as control variables for use in our regression analyses of dividend announcements covering the same time frame above.

These attributes include market capitalization, ROA, and leverage ratios. Once this information was gathered from multiple sources, it was combined into one comprehensive dataset containing both the announcement of dividends with their related return as well as stock performance within their respective markets and firm characteristic data.

3.4 Event Study Methodology

The stock market response to dividend announcements is the subject of examination using an event study technique in order to assess its behavior. Within an event study framework, the abnormal return of a stock through time (on a particular date) is determined using the actual return of that stock and the predicted return for that stock had the dividend announcement not occurred (MacKinlay, 1997). The following are the structured pieces that make up the event study framework to evaluate abnormal stock returns.

3.4.1 Stock Returns

Daily stock returns are calculated using adjusted closing prices as follows:

$$R_{it} = \frac{P_{it} - P_{it-1}}{P_{it-1}}$$

Where:

R_{it} = return of stock i on day t

P_{it} = adjusted closing price of stock i on day t

P_{it-1} = adjusted closing price of stock i on day t-1

Adjusted prices are used to account for stock splits and dividend adjustments, ensuring accurate return calculations

3.4.2 Event date (Announcement Date)

The event date ($t = 0$) is identified as the date that the company announces its dividend decision publicly through a press release. This specific date is considered the point in time that the market will receive new information regarding the dividend decision. In a semi-strong efficient market, any positive or negative price reaction to a dividend announcement is anticipated to occur at or near the event date.

3.4.3 Estimation Window

The estimation window of -250 to -30 trading days relative to the date of the event, spanning 220 trading days, provides a statistically reliable estimate of the relationship between individual stock returns and market returns without overlapping with the event window to avoid contamination due to the announcement itself. The estimation window is also used as a basis for estimating parameters (α) and (β) of the expected return model to determine the normal return-generating process for each stock.

3.4.4 Event Window

Within the event window, we examine how quickly the market reacts to dividend announcements. Short-term event windows are used, including (-1, +1) and (-3, +3) trading days, to determine both quick and delayed reactions to these announcements.

Two event windows are examined in this study. The short window of three trading days (-1, +1) is the most widely adopted specification in the dividend announcement literature, following Aharony and Swary (1980). The day before the announcement ($t = -1$) is included to capture any partial pre-announcement price movement, while the day after ($t = +1$) accounts for the common practice among S&P 500 firms of releasing dividend decisions after the market close, which defers the primary price reaction to the following morning.

The wider window of seven trading days (-3, +3) serves a complementary purpose, it tests whether systematic pre-announcement drift or post-announcement price adjustment exists beyond the core three-day window, and provides a buffer against potential inaccuracies in the recorded announcement date.

Longer event windows, such as (-5, +5) or beyond, were considered but rejected on two grounds. First, wider windows increase the likelihood that other firm-specific events like earnings announcements, share buybacks, or management changes occur within the window and may contaminate the abnormal return estimate. Second, longer windows accumulate additional noise in the cumulative abnormal return, reducing the statistical power of the hypothesis tests.

The two windows selected therefore reflect the standard approach in the event study literature and are appropriate for isolating the market reaction specifically attributable to the dividend announcement.

3.4.5 Expected Return Model

The expected return has been computed using the market model. The relationship between returns of the particular firms' stock and market portfolio is defined as linear. Thus, the market model can be expressed as follows:

$$R_{it} = \alpha_i + \beta_i R_{mt} + \epsilon_{it}$$

Where

R_{it} represents the return on stock i at time t

R_{mt} denotes the return on the market index (S&P 500)

ϵ_{it} is the error term

α_i and β_i are estimated over the estimation window using ordinary least squares (OLS).

The market model is widely used in event studies due to its ability to control for market-wide movements and improve the precision of abnormal return estimates (MacKinlay, 1997).

3.4.6 Abnormal Return

Abnormal return is defined as the difference between a security's actual (observed) return and its expected return, where expected returns are estimated from general movements within the overall financial market. Therefore, abnormal returns reflect the portion of change in price of an individual security that cannot be explained by events within the overall equity market and that can be attributed instead to changes in the firm's value, such as dividend declarations or other company-specific events.

Once expected returns are estimated, abnormal returns (AR) are calculated as the difference between actual returns and expected returns:

$$AR_{it} = R_{it} - (\alpha_i + \beta_i R_{mt})$$

Abnormal returns represent the portion of stock returns that cannot be explained by general market movements and are therefore attributed to firm-specific events, such as dividend announcements. Positive abnormal returns indicate that the market reacts favorably to the announcement, while negative abnormal returns suggest an unfavorable reaction.

Event study methodology frequently relies on the use of abnormal returns to determine the extent to which an announcement or disclosure of specific event had an influence on the price of an individual equity security (MacKinlay, 1997). The aggregated abnormal returns over an event window define the cumulative abnormal return (CAR) for the same window, providing a direct measure of how the market reacted to the announcement/market event. Statistically significant CARs indicate that the market perceives value-relevant information in dividend declarations and reconsideration of the value of the firm as a result of this value-relevant information.

3.4.7 Cumulative Abnormal Return

In order to summarize the complete market reaction across the event window, it is necessary to gather up the abnormal returns into a total abnormal return called cumulative abnormal return (CAR):

$$CAR_{(t_1, t_2)} = \sum_{t=t_1}^{t_2} AR_{it}$$

Where t_1 and t_2 are the start and end days of your event window (e.g.-1 to +1, or-3 to +3).

CAR is an aggregate statistic that measures all the effects of the dividend announcement on the price of the stocks for a particular event period. A cumulative abnormal return can be regarded as giving a ground for determining if the dividend announcement contained relevant information to investors such that they revised their estimates of the firm's value. More generally, this event study methodology provides a structural process to evaluate how dividend announcements impact stock prices. By separating the abnormal returns within the designated event window, the methodology also allows for identifying market reactions resulting from the information related to dividend announcements. Consequently, this methodology answers the study's primary research question.

3.4.8 Average Abnormal Return (AAR)

To measure the average market reaction across all firms on each day in the event window, average abnormal returns (AAR) are calculated as follows:

$$AAR_t = \frac{1}{N} \sum_{i=1}^N AR_{it}$$

Where:

AAR_t = average abnormal return on day t

N = number of firms in the sample

AR_{it} = abnormal return for firm i on day t

Average abnormal returns show the average daily market reaction to dividend announcements across all firms.

3.4.9 Cumulative Average Abnormal Return (CAAR)

To measure the overall market reaction across all firms over the event window, cumulative average abnormal returns (CAAR) are calculated as:

$$CAAR_{(t_1, t_2)} = \sum_{t=t_1}^{t_2} AAR_t$$

CAAR provides an overall measure of the market reaction to dividend announcements across all firms in the sample.

3.4.10 Event Study Variables

The event study methodology relies on several key variables used to estimate expected returns, calculate abnormal returns (AR), and measure the stock market reaction to dividend announcements. These variables are distinct from the explanatory variables employed in the subsequent cross-sectional regression analysis and are used solely for the purpose of estimating abnormal performance around the event date.

Table 1: Definition of Event Study Variables

Symbol	Definition	Description
R_{it}	Actual Stock Return	Daily return of firm i on trading day t
R_{mt}	Market Return	Daily return on the S&P 500 index on trading day t
α_i	Intercept Term	Firm-specific intercept estimated from the market model
β_i	Beta Coefficient	Firm-specific sensitivity of stock returns to market returns
ϵ_{it}	Error Term	Residual component of stock returns not explained by the market model

Using these inputs, the study derives several abnormal return measures to evaluate the stock market reaction to dividend announcements.

3.4.11 Statistical Tests

To determine whether abnormal returns are statistically different from zero, t-tests are conducted for average abnormal returns and cumulative average abnormal returns.

The test statistic is calculated as:

$$t = \frac{AAR_t}{SE(AAR_t)} \text{ and } t = \frac{CAAR}{SE(CAAR)}$$

Where $SE(AAR_t)$ is the standard error of the average abnormal return on day t and $SE(CAAR)$ is the standard error of the cumulative average abnormal return over the event window.

Statistical significance is evaluated at the 1%, 5%, and 10% significance levels. If abnormal returns are significantly different from zero, it indicates that dividend announcements contain value-relevant information and that the stock market reacts to dividend announcements.

3.5 Cross Sectional Analysis

While the event study methodology establishes whether dividend announcements generate statistically significant abnormal returns on average, it does not explain why the magnitude of market reactions differs across dividend announcement events. To investigate the determinants of cross-sectional variation in abnormal returns, a subsequent regression analysis is conducted. This second-stage analysis examines whether different types of dividend announcements and firm-specific characteristics explain variation in cumulative abnormal returns across events.

3.5.1 Cross Sectional Regression Analysis

A cross-sectional regression is performed to test the hypotheses of this study using the cumulative abnormal return (CAR) as the dependent variable. This enables an assessment of whether different types of dividend announcements generate distinct and statistically significant market reactions.

Since every dividend announcement in the sample belongs to one of three mutually exclusive categories dividend increase, dividend decrease, or dividend initiation including a dummy variable for all three categories simultaneously would cause perfect multicollinearity with the intercept term, making the model impossible to estimate. To avoid this, dividend decrease announcements are used as the reference category, and only two dummy variables are included in the regression. All coefficients are therefore interpreted relative to the market reaction associated with dividend decreases.

The regression model is specified as follows:

$$CAR_i = \beta_0 + \beta_1 DividendIncrease_i + \beta_2 DividendInitiation_i + \beta_3 Controls_i + \epsilon_i$$

Where CAR_i is the cumulative abnormal return for firm i over the specified event window. $DividendIncrease_i$ and $DividendInitiation_i$ are dummy variables indicating the type of dividend announcement, with $DividendDecrease_i$ serving as the omitted reference category. The vector $Controls_i$ includes firm-specific characteristics such as firm size, profitability, leverage, and valuation.

The coefficients β_1 and β_2 capture the average market reaction to dividend increases and dividend initiations respectively, each measured relative to the market reaction for dividend decreases. A positive and statistically significant coefficient indicates that the market reacts more favorably to that announcement type compared to dividend decreases. The intercept β_0 represents the baseline average CAR associated with dividend decrease announcements.

3.5.2 Dependent Variable

The dependent variable in the cross-sectional regression is the cumulative abnormal return (CAR) estimated from the event study methodology described in Section 3.4. CAR measures the aggregate abnormal stock return over the selected event window surrounding the dividend announcement and serves as a proxy for the market's valuation response to the announcement.

Using CAR as the dependent variable is appropriate because it captures the total abnormal return attributable to the dividend announcement over the event window, thereby accounting for possible information leakage prior to the announcement and delayed market adjustment following the announcement.

3.5.3 Independent Variables

The primary explanatory variables of interest are dummy variables representing the type of dividend announcement. As noted above, dividend decrease is the omitted reference category against which the other announcement types are compared.

Dividend Increase

A binary variable equal to 1 if the dividend announcement represents an increase relative to the firm's previous dividend payment, and 0 otherwise. Under dividend signaling theory, dividend increases are expected to generate positive abnormal returns relative to dividend decreases, as they signal management's confidence in future earnings and cash flow sustainability.

Dividend Decrease (Reference Category)

Dividend decrease announcements serve as the baseline reference category and are therefore not included as a separate dummy variable in the regression. The intercept term β_0 captures the average CAR associated with dividend decrease events. This choice is theoretically motivated, as dividend decreases represent the negative signal against which the positive signals of increases and initiations are compared.

Dividend Initiation

A binary variable equal to 1 if the dividend announcement represents the first-time initiation of dividend payments by the firm, and 0 otherwise. Dividend initiations are expected to generate positive abnormal returns relative to dividend decreases, as they may signal improved financial maturity and confidence in sustainable future cash flows.

3.5.4 Control Variables

To account for heterogeneity across firms, the analysis includes a set of control variables that capture key firm characteristics. These variables are incorporated into the regression model to ensure that the estimated effects of dividend announcements are not confounded by underlying differences in firm size, financial performance, or risk profile. The selection of these control variables is guided by dividend signaling theory, the free cash flow hypothesis, and established empirical practice in event study literature, as detailed below for each variable.

The first control variable is firm size which is assessed using market capitalization as a proxy. Firm size is measured as the natural logarithm of market capitalization:

$$Firm\ Size = \ln(Market\ Capitalisation)$$

Larger firms tend to have more stable earnings, greater analyst coverage, and higher liquidity, which may reduce the magnitude of stock price reactions to dividend announcements. Following Fama and French (1993) and Grullon et al. (2002), firm size is included because information asymmetry is inversely related to firm size, meaning dividend announcements from smaller firms are expected to carry a stronger signal, producing larger abnormal returns.

The second control variable is profitability which uses return on assets (ROA) as a proxy for measuring profitability for a firm; additionally, this indicates how well a firm generates earnings from its assets. A more profitable company might be viewed as more capable of continuing to pay dividends, and in this way could affect how investors react to dividend announcements. Consistent with the free cash flow hypothesis (Jensen, 1986), profitability is included because firms with higher earnings have greater capacity to sustain dividends, so the market reaction to a dividend announcement may differ based on the firm's ability to maintain it.

$$ROA = \frac{Net\ Income}{Total\ Assets}$$

The amount of leverage measured using the debt-to-equity ratio is a third control variable. This measurement reflects the company's financial risk. Companies with high leverage will face increased financial limitations, which potentially impacts how investors will interpret dividend announcements. Following the agency cost framework (Jensen and Meckling, 1976), leverage is included because highly leveraged firms face greater financial constraints, and dividend announcements by such firms may be interpreted more skeptically by the market, dampening the abnormal return

$$Leverage = \frac{Total\ Debt}{Total\ Equity}$$

In addition to these three control variables, valuation as reflected through the price to book ratio has been included in order to allow for industry differences in terms of growth options and/or market expectations among firms. The price-to-book ratio is included as a proxy for growth opportunities (Fama and French, 1993; Lang and Litzenberger, 1989). Firms with high P/B ratios are considered growth firms and their dividend announcements may generate different market reactions compared to value firms, as dividends from growth firms reduce funds available for investment.

$$P/B = \frac{Market\ Value\ Of\ Equity}{Book\ Value\ Of\ Equity}$$

Lastly, industry classification using typically sector based dummy variable to account for industry specific factors that may affect dividend behaviour and share price movement will be included. Industry fixed effects are included to control for systematic differences in dividend policy and investor expectations across sectors, consistent with the approach used by Michaely, Thaler and Womack (1995). Without this control, sector-level confounders could bias the estimated coefficients on the dividend type variables.

The result is by using these control variables, it strengthens the robustness of the empirical analysis to support that what has been observed through the stock market reaction can be attributed to the dividend announcements and not due to other firm specific variables.

Expected Signs of Coefficients

Based on dividend signaling theory and prior empirical literature, the expected signs of the regression coefficients are as follows:

Table 2: Expected Signs of Regression Coefficients

Variable	Expected Sign
Dividend Increase	+
Dividend Decrease	Reference Category
Dividend Initiation	+
Firm Size (Log Market Capitalization)	-
Return on Assets (ROA)	+
Leverage (Debt-to-Equity)	-
Price-to-Book Ratio	+

Dividend increases and dividend initiations are expected to generate positive abnormal returns because they signal strong future earnings and financial stability. Dividend decreases are expected to generate negative abnormal returns because they may signal financial difficulties or declining profitability.

Firm size is expected to have a negative relationship with abnormal returns because larger firms typically have more information available to investors, reducing the magnitude of announcement effects. Profitability is expected to have a positive relationship with abnormal returns, while leverage is expected to have a negative relationship due to increased financial risk. Firms with higher price-to-book ratios may experience stronger market reactions due to growth expectations.

3.6 Statistical Tests

Statistical tests are used, primarily t-tests, to determine if abnormal returns are significantly different from zero for dividends announced during the study period.

In addition, they performed significance testing at traditional confidence levels (1%, 5%, and 10% levels) to evaluate the reliability of their estimated coefficients (the value of abnormal return) and test statistics (also defined as t-test statistic). In addition to the overall findings being statistically significant or not, when abnormal returns are significantly different from zero there is a good reason to conclude that they are due to the events studied and, therefore, that the stock market has likely reacted to the dividend announcement made by the company.

Also, we compared the returns between companies announcing different types of dividend events; i.e., those announcing increases vs decreases, or initiating dividends. Performing these comparisons allows the us to determine any asymmetry in the reaction by investors when responding to positive or negative dividends, and to understand whether they reacted differently to newly initiated dividends. All of the statistical tests used in this study give support for conducting a thorough analysis of the impact that dividend announcements have had on the stock market.

3.6.1 Test of Average Abnormal Returns (AAR)

The statistical significance of average abnormal returns (AAR) for each day in the event window is tested using the t-statistic:

$$t(AAR_t) = \frac{AAR_t}{SE(AAR_t)}$$

Where:

AAR_t = average abnormal return on day t

$SE(AAR_t)$ = standard error of average abnormal returns

If the t-statistic is statistically significant, it indicates that the average abnormal return on that day is significantly different from zero.

3.6.2 Test of Cumulative Average Abnormal Returns (CAAR)

To evaluate the overall market reaction over the event window, cumulative average abnormal returns (CAAR) are tested using the following t-statistic:

$$t(CAAR) = \frac{CAAR}{SE(CAAR)}$$

Where:

$CAAR$ = Cumulative average abnormal returns over the event window

$SE(CAAR)$ = Standard error of cumulative average abnormal returns

If CAAR is statistically significant, it indicates that dividend announcements generate a significant overall market reaction over the event window.

3.6.3 Significance Levels

Statistical significance is evaluated using conventional significance levels commonly used in financial research:

Table 3: Statistical Significance Levels

Significance Level	Interpretation
1%	Highly significant
5%	Statistically significant
10%	Weakly significant

A p-value less than 0.01 indicates strong evidence that abnormal returns are different from zero. A p-value less than 0.05 indicates statistical significance, while a p-value less than 0.10 indicates weak statistical significance.

3.6.4 Regression Statistical Tests

In the regression analysis, t-tests are also used to evaluate whether the regression coefficients are statistically different from zero. The t-statistic for each regression coefficient is calculated as follows:

$$t(\beta_j) = \frac{\beta_j}{SE(\beta_j)}$$

Where:

β_j = estimated regression coefficient

$SE(\beta_j)$ = Standard error of the coefficient

If a regression coefficient is statistically significant, it indicates that the corresponding variable has a significant effect on cumulative abnormal returns.

3.7 Summary of Methodology

The outline of the empirical framework outlined in this chapter provides a means of assessing how the stock market reacted to dividend announcements. This study is based on a quantitative research design, using an event study methodology to measure the abnormal return on stock within the defined window of the announcement (MacKinlay, 1997) for firms identified as part of the S&P 500.

The firm that is the subject of this research was traced from reputable financial databases such as FactSet. Types of data used include information related to dividend announcements, stock returns, market index data, and characteristics related to the firms being examined.

The analysis will be done in two stages.

Firstly, the event study methodology is used to calculate abnormal returns (AR), average abnormal returns (AAR), cumulative abnormal returns (CAR), and cumulative average abnormal returns (CAAR) around dividend announcement dates using the market model to estimate expected returns. These measures allow for the identification of stock price reactions to dividend announcements.

Secondly, a regression analysis on the various types of dividend announcements (i.e, increases, decreases and initiations), controlling for firm-level characteristics, to investigate whether there is a distinguishable difference in market reaction across different types of dividends. By utilizing these two methods, I hope to provide a structured and empirical approach to assess how markets will respond to dividend announcements in today's financial environment.

CHAPTER 4

EMPIRICAL RESULTS

This chapter presents the empirical findings of the study examining stock market reactions to dividend announcements made by firms included in the S&P 500 index over the period January 2011 to December 2025. The analysis proceeds in four stages.

First, Section 4.1 describes the final sample composition and provides descriptive statistics for the distribution of cumulative abnormal returns (CARs) across the three announcement categories. Second, Section 4.2 presents the event study results, reporting average abnormal returns (AARs) and cumulative average abnormal returns (CAARs) around the announcement date for the full sample and for each announcement type, followed by formal hypothesis tests on mean CARs and a summary of the event study findings.

Third, Section 4.3 reports a set of diagnostic tests performed after the event study, including normality assessments, multicollinearity checks, and correlation analysis, which validate the statistical choices made in the cross-sectional regression. Fourth, Section 4.4 presents the cross-sectional regression results, which examine whether announcement type and firm-level characteristics explain variation in CARs across events.

All event study estimates are based on the market model, with parameters estimated over a 220-trading-day window ending 30 days prior to each announcement. Two event windows are examined throughout: a short window of three trading days centred on the announcement date (-1, +1) and a wider window of seven trading days (-3, +3), which captures any pre-announcement information leakage and post-announcement price adjustment. Statistical significance is evaluated at the 1%, 5%, and 10% levels, and HC3 heteroskedasticity-robust standard errors are applied in all regression specifications, as motivated by the normality diagnostics reported in Section 4.3.

4.1 Sample Description and Descriptive Statistics

The sample is drawn from ordinary dividend announcements made by firms that were constituents of the S&P 500 index as of 2 February 2025, covering the 15-year period from 1 January 2011 to 31 December 2025. A total of 22,118 ordinary dividend announcements were identified across 414 firms.

Each announcement was classified relative to the firm's immediately preceding ordinary dividend payment of the same frequency into one of four mutually exclusive categories. An announcement is classified as a dividend increase if the declared dividend per share (DPS) is strictly greater than the preceding DPS. An announcement is classified as a dividend decrease if the current DPS is strictly less than the preceding DPS. An announcement

is classified as a dividend initiation if the firm has no prior ordinary dividend payment on record within the sample period, indicating the first regular dividend payment by that firm. An announcement is classified as no change if the current DPS is exactly equal to the preceding DPS.

Events classified as no change where the dividend per share was identical to the prior payment were excluded from the analysis, as they carry no incremental signal under dividend signaling theory. This exclusion removed 16,881 events, leaving a raw pool of 5,237 events across the three analytical categories.

Daily adjusted closing prices used to compute stock returns span from 31 December 2009 to 31 December 2025, with the earlier start date ensuring that sufficient price history is available to construct the estimation window for announcement events occurring at the beginning of the study period

A minimum of 60 trading days with valid recorded price observations is required within the estimation window ($t-250$ to $t-30$) for an event to be included in the analysis. This count is based exclusively on days on which the stock actually traded and a return could be computed; calendar days, public holidays, and days with missing or suspended price data do not count toward the threshold.

The market model is estimated by ordinary least squares over the $[t-250, t-30]$ window, which spans 220 trading days and provides the full estimation period available to each event (MacKinlay, 1997). The 60-observation requirement is therefore not the length of the estimation window itself, but a minimum-data floor that an event must meet to be retained: it is the smallest number of valid daily returns within that window for which the OLS estimates of alpha and beta can be regarded as sufficiently stable.

The 60-day floor reflects a deliberate balance between estimation reliability and sample retention. A lower threshold, such as 30 or 40 trading days, was considered but rejected. With so few observations, the OLS regression produces unstable estimates of alpha and beta with large standard errors, and this imprecision feeds directly into the abnormal return calculations, introducing noise into the results. A substantially higher threshold, such as 80 or 100 days, was also rejected. Because the data requirement is applied at the individual-event rather than the firm level, a stricter floor would disproportionately exclude events from firms with shorter listing histories, recent spin-offs, or trading gaps caused by corporate restructuring, reducing the sample without a corresponding improvement in estimation quality. The 60-observation requirement retains these events while still ensuring that each retained event has enough data for reliable parameter estimation.

The 60-day minimum therefore strikes the right balance between maintaining estimation reliability and retaining the broadest possible set of usable events.

Events failing this criterion were dropped individually rather than firm-wide, preserving the maximum number of usable observations. After applying this filter, the final event study sample consists of 5,237 events for the short event window (-1, +1) and 5,238 events for the long event window (-3, +3). Table 4.1 provides a full summary of the sample construction.

Table 4.1 : Sample Summary

Item	Value
Study Period	1 January 2011 to 31 December 2025
Market Index	S&P 500 Index
Total Ordinary Announcements (2011–2025)	22,118
Unique Firms	414
Dividend Increases (H1)	4,614
Dividend Decreases (H2)	244
Dividend Initiations (H3)	379
No Change (excluded)	16,881
Events retained- Window (-1, +1)	5,237
Events retained- Window (-3, +3)	5,238
Estimation Window	t-250 to t-30 (220 trading days)
Short Event Window	(-1, +1) trading days
Long Event Window	(-3, +3) trading days
Minimum Estimation Observations	60
Expected Return Model	OLS Market Model (S&P 500 as market proxy)

Note: The total analysis sample reflects events with sufficient data in the estimation window. The minor difference between the (-1,+1) and (-3,+3) samples reflects marginal data availability around individual event boundaries. Firm count reflects S&P 500 constituents as of 2 February 2025.

Dividend increases constitute 4,614 retained events in the short window sample. This dominance is consistent with the established finding that S&P 500 firms tend to smooth dividends upward over time and are reluctant to reduce payouts (Lintner, 1956; Brav et al., 2005). Dividend decreases are comparatively rare, with only 244 events retained in the final sample, reflecting the well-documented managerial tendency to avoid cuts even under financial pressure. Dividend initiations number 379 retained events, representing instances where a firm began paying ordinary dividends for the first time within the sample period.

4.1.1 Descriptive Statistics of Cumulative Abnormal Returns

Table 4.2 reports descriptive statistics for the distribution of cumulative abnormal returns (CARs) across announcement types and both event windows. All values are expressed as percentages.

Table 4.2: Descriptive Statistics of CAR by Announcement Type and Event Window

Type	Window	N	Mean (%)	Median (%)	SD (%)	Min (%)	Max (%)	Skewness	Ex. Kurtosis
All Events	(-1,+1)	5,235	0.0845	0.0833	5.2019	-100.0000	45.0373	-8.8019	169.5546
Decrease	(-1,+1)	244	-0.4085	-0.2309	8.4741	-98.3378	40.5219	-5.9439	74.3052
Increase	(-1,+1)	4,614	0.1316	0.1335	5.1166	-100.0000	45.0373	-9.0356	175.3365
Initiation	(-1,+1)	379	-0.1719	-0.3456	2.9881	-24.7061	13.3208	-1.5521	15.2067
All Events	(-3,+3)	5,238	0.2298	0.1368	7.3946	-181.8471	150.3162	-5.5089	179.3140
Decrease	(-3,+3)	244	-0.6014	-0.3499	15.7648	-181.8471	70.7080	-6.4957	77.3257
Increase	(-3,+3)	4,615	0.2688	0.2037	6.5428	-149.3166	70.7080	-6.7493	130.1691
Initiation	(-3,+3)	379	0.2896	-0.0916	8.6492	-28.0186	150.3162	13.7199	237.5787

Note: CAR values are expressed as percentages. Kurtosis is excess kurtosis, where a value of zero corresponds to the normal distribution. All Events pools dividend increases, decreases, and initiations. N reflects events with sufficient data in the estimation window.

Across the full sample, the mean CAR over the short window (-1, +1) is 0.08%, with a median of 0.08%, indicating a modest but positive central tendency alongside substantial dispersion (SD = 5.20%). Over the wider (-3, +3) window, the mean CAR rises to 0.23%, suggesting that a portion of the price adjustment occurs in the days surrounding the core announcement date rather than at $t = 0$ alone.

When disaggregated by announcement type, several important patterns emerge. Dividend increases display a mean CAR of 0.13% over the (-1, +1) window and 0.27% over the (-3, +3) window, consistent with the prediction under H1 that increases generate positive abnormal returns. The positive mean is reinforced by near-identical median values (0.13% and 0.20% respectively), suggesting the central tendency is not driven by outliers (it was not just a handful of extreme reactions influencing average).

Dividend decreases exhibit a negative mean CAR of -0.41% over (-1, +1) and -0.60% over (-3, +3), consistent with the prediction under H2. The considerably larger standard deviation for decreases (8.47% over (-1, +1)) compared to increases (5.12%) reflects greater heterogeneity in market reactions to negative dividend signals, which may reflect variation in the severity and context of individual cuts.

Dividend initiations show a slightly negative mean CAR of -0.17% over (-1, +1) but a positive mean of 0.29% over the wider (-3, +3) window. The sign reversal between windows warrants careful interpretation. Over the short window, the announcement date itself appears to generate a negative reaction on average, which contrasts with the prediction under H3. However, the (-3, +3) result suggests that positive pre-announcement drift or post-announcement adjustment may shift the cumulative return into positive territory.

This pattern may also partly reflect the fact that the initiation subsample is relatively small at 379 events. With a small group, a few extreme reactions can pull the average in one direction and make it less representative of what typically happens. This is supported by the data, the excess kurtosis for initiations over the (-3, +3) window is 237.58, which is extremely high, and the skewness is +13.72, meaning the distribution has a long right tail driven by a small number of very large positive returns.

Across all groups and both windows, the CAR distributions show clear departures from normality. The skewness and excess kurtosis values are far beyond what would be expected under a normal distribution, which simply means the returns are not evenly spread and are heavily influenced by extreme events at both ends. These characteristics are tested formally in Section 4.3 and are the reason why robust standard errors are used throughout the regression analysis rather than relying on standard methods that assume normality. The visual distribution of CARs is shown in Appendix 2.

4.2 Event Study Results

This section presents the results from the event study analysis. The first part looks at how stock returns moved on each day within the event window, both for the overall sample and for each type of dividend announcement. The second part tests whether the cumulative abnormal returns are statistically different from zero, which allows us to evaluate the three hypotheses set out in Chapter 3.

4.2.1 Average Abnormal Returns and Cumulative Average Abnormal Returns

Tables 4.3a to 4.3d report the daily AARs and CAARs from day -3 to day +3 around the announcement date, presented separately for the full sample and each announcement subsample. The significance levels in each table refer to the daily t-test of $H_0: AAR = 0$. Figures 4.1 and 4.2 provide a visual summary of these patterns.

Table 4.3a: Daily AAR and CAAR- Full Sample

Day	N	AAR (%)	SE AAR (%)	t(AAR)	p-value	CAAR (%)	Sig.
-3	5,238	0.0127	0.0374	0.339	0.7344	0.0127	
-2	5,238	0.0589	0.0284	2.076	0.0379	0.0716	**
-1	5,237	0.0151	0.0368	0.412	0.6806	0.0867	

0	5,237	-0.0846	0.0439	-1.928	0.0540	0.0021	*
+1	5,237	0.1540	0.0492	3.130	0.0018	0.1561	***
+2	5,237	0.0646	0.0302	2.136	0.0327	0.2206	**
+3	5,237	0.0092	0.0333	0.277	0.7822	0.2298	

Note: AAR = Average Abnormal Return. CAAR = Cumulative Average Abnormal Return accumulated from day -3 to day t. SE AAR = standard error of AAR. $t(AAR)$ = test statistic for $H_0: AAR = 0$ on each trading day, evaluated using a one-sample t-test. Significance levels refer to this daily t-test: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All values in percentages. N = number of events per day.

The full sample results show limited pre-announcement price movement, with only day -2 producing a marginally significant AAR of 0.06% ($t = 2.08$, $p = 0.038$). The announcement day itself (day 0) yields a weakly significant negative AAR of -0.08% ($p = 0.054$), likely reflecting the common practice among S&P 500 firms of releasing dividend decisions after market close, which defers the price reaction to the following day. Day +1 records the strongest reaction in the full sample at 0.15% ($t = 3.13$, $p = 0.002$), followed by a smaller positive AAR on day +2 (0.06%, $p = 0.033$). The CAAR reaches 0.23% by day +3, confirming a small but positive overall market reaction to dividend announcements.

Table 4.3b: Daily AAR and CAAR- Dividend Decrease Subsample

Day	N	AAR (%)	SE AAR (%)	t(AAR)	p-value	CAAR (%)	Sig.
-3	244	-0.3584	0.4136	-0.867	0.3871	-0.3584	
-2	244	0.2814	0.1392	2.022	0.0443	-0.0770	**
-1	244	0.2036	0.1406	1.449	0.1488	0.1266	
0	244	-0.2426	0.1840	-1.319	0.1885	-0.1160	
+1	244	-0.3695	0.4683	-0.789	0.4309	-0.4855	
+2	244	0.2288	0.1430	1.600	0.1108	-0.2567	
+3	244	-0.3447	0.4169	-0.827	0.4091	-0.6014	

Note: AAR = Average Abnormal Return. CAAR = Cumulative Average Abnormal Return accumulated from day -3 to day t. SE AAR = standard error of AAR. $t(AAR)$ = test statistic for $H_0: AAR = 0$ on each trading day, evaluated using a one-sample t-test. Significance levels refer to this daily t-test: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All values in percentages. N = number of events per day.

The decrease subsample ($N = 244$) shows an inconsistent daily pattern with no statistically significant AAR at conventional levels across the window, except for an unexpected positive return on day -2 (0.28%, $p = 0.044$), which is likely attributable to noise given the small sample size and high dispersion within this group. More informative is the CAAR side, which trends persistently negative from -0.36% on day -3 to -0.60% by day +3. The steady decline in returns over the window suggests that some investors may have already picked up on signs that the company was in financial difficulty before the dividend cut was officially announced. In other words, the bad news was gradually being reflected in the stock price in the days leading up to the announcement rather than causing a sudden sharp reaction on a single day

Table 4.3c: Daily AAR and CAAR- Dividend Increase Subsample

Day	N	AAR (%)	SE AAR (%)	t(AAR)	p-value	CAAR (%)	Sig.
-3	4,615	0.0171	0.0352	0.486	0.6269	0.0171	
-2	4,615	0.0432	0.0298	1.449	0.1474	0.0603	
-1	4,614	-0.0124	0.0400	-0.311	0.7559	0.0478	
0	4,614	-0.0474	0.0472	-1.003	0.3158	0.0005	
+1	4,614	0.1914	0.0475	4.030	0.0001	0.1919	***
+2	4,614	0.0506	0.0319	1.588	0.1123	0.2425	
+3	4,614	0.0264	0.0293	0.901	0.3678	0.2689	

Note: AAR = Average Abnormal Return. CAAR = Cumulative Average Abnormal Return accumulated from day -3 to day t. SE AAR = standard error of AAR. $t(AAR)$ = test statistic for $H_0: AAR = 0$ on each trading day, evaluated using a one-sample t-test. Significance levels refer to this daily t-test: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All values in percentages. N = number of events per day.

The increase subsample (N = 4,614) produces the clearest pattern. The pre-announcement CAAR is essentially flat at 0.05% by day -1, with no significant daily AARs, indicating that investors are not anticipating increases ahead of disclosure. The reaction concentrates almost entirely on day +1, where the AAR of 0.19% is the most precisely estimated result in the entire analysis ($t = 4.03$, $p < 0.001$). The CAAR then stabilizes at 0.27% through day +3, with no further drift. This rapid single-day adjustment followed by price stability is consistent with semi-strong form market efficiency.

Table 4.3d: Daily AAR and CAAR- Dividend Initiation Subsample

Day	N	AAR (%)	SE AAR (%)	t(AAR)	p-value	CAAR (%)	Sig.
-3	379	0.1978	0.1122	1.763	0.0787	0.1978	*
-2	379	0.1071	0.1184	0.904	0.3666	0.3049	
-1	379	0.2296	0.1139	2.016	0.0445	0.5344	**
0	379	-0.4366	0.1528	-2.857	0.0045	0.0979	***
+1	379	0.0351	0.1917	0.183	0.8547	0.1330	
+2	379	0.1287	0.1245	1.034	0.3018	0.2617	
+3	379	0.0279	0.1118	0.250	0.8030	0.2896	

Note: AAR = Average Abnormal Return. CAAR = Cumulative Average Abnormal Return accumulated from day -3 to day t. SE AAR = standard error of AAR. $t(AAR)$ = test statistic for $H_0: AAR = 0$ on each trading day, evaluated using a one-sample t-test. Significance levels refer to this daily t-test: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All values in percentages. N = number of events per day.

The initiation subsample ($N = 379$) exhibits the most complex pattern. The CAAR builds steadily to 0.53% by day -1, supported by a marginally significant AAR on day -3 (0.20%, $p = 0.079$) and a significant AAR on day -1 (0.23%, $p = 0.045$), suggesting that some market participants anticipate the initiation ahead of its disclosure. However, on day 0, the AAR reverses sharply to -0.44% ($t = -2.86$, $p = 0.005$), the largest single-day negative reaction in the entire analysis. This reversal is consistent with a sell-the-news effect, where investors who bought shares in anticipation of good news liquidate once the announcement is confirmed, or alternatively with the announcement falling short of already elevated expectations. The CAAR subsequently recovers to 0.29% by day +3, though as discussed in Section 4.2.2, this does not reach statistical significance.

Figure 4.1: Cumulative Average Abnormal Returns (CAAR) by Announcement Type Event Window (-3, +3)

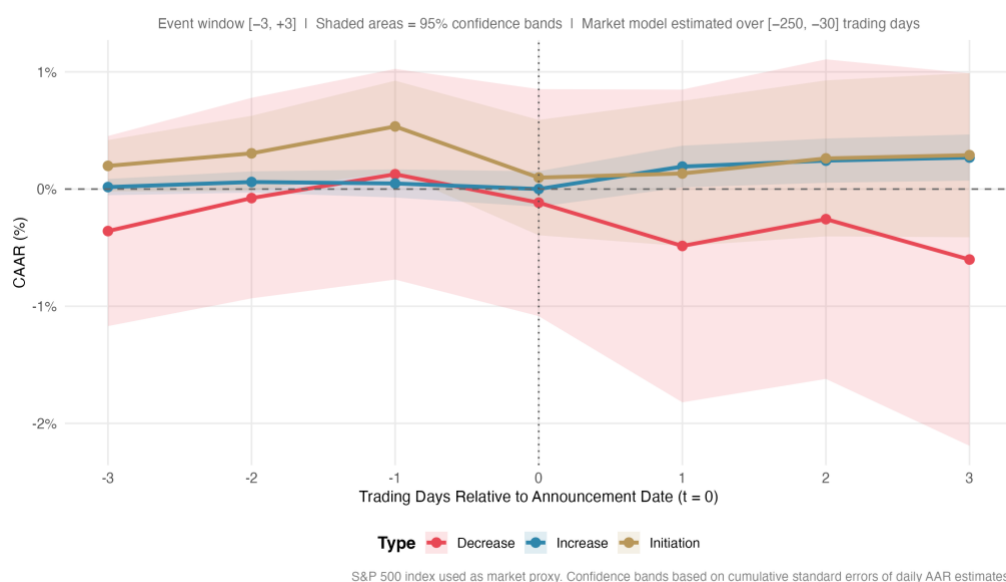


Figure 4.1 shows the CAAR for each announcement type plotted across the seven-day window. The three lines move in clearly different directions, which makes the chart easy to read. The red line for dividend decreases moves downward throughout the window with no sign of recovery, finishing at -0.60% by day +3. The blue line for dividend increases stays flat before the announcement and then rises sharply after day 0, before levelling off a clean and decisive reaction. The gold line for dividend initiations is the most interesting; it climbs to a peak of around 0.53% on day -1, then drops sharply at day 0, before slowly recovering. This confirms the build-up and reversal pattern described above.

Figure 4.2: Average Abnormal Returns (AAR) by Day and Announcement Type Event Window (-3, +3)

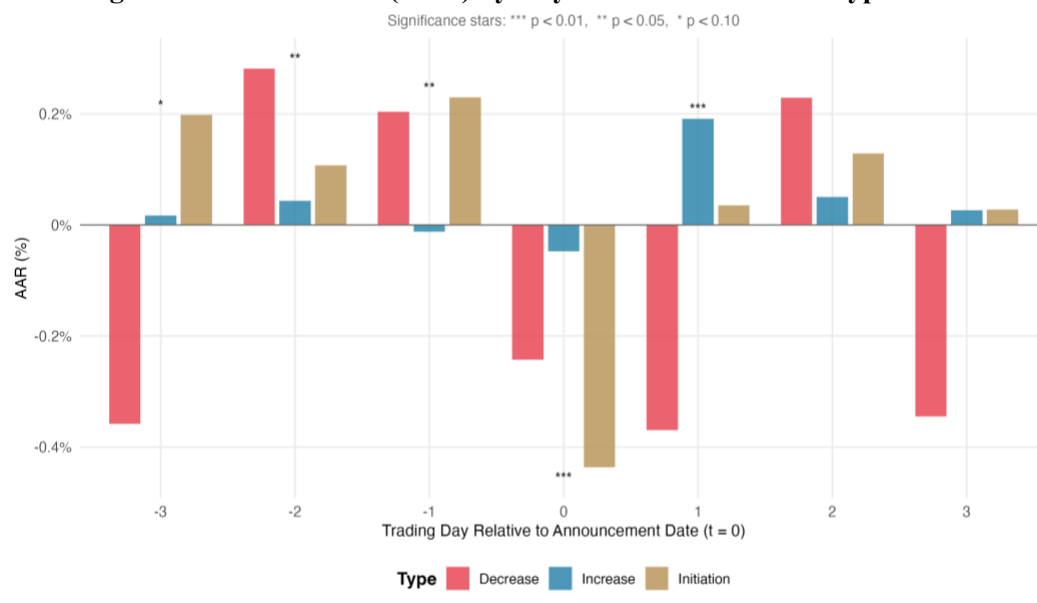


Figure 4.2 shows the same information as a bar chart, with each bar representing the average abnormal return on a given day for each announcement type. The chart makes it easy to see which days drive the reactions. For increases, the tallest bar is clearly at day +1, confirming that almost all of the market reaction happens on a single day. For decreases, the bars are mixed in sign with no single dominant day, which is consistent with the noisy and unclear reaction described earlier. For initiations, the large negative bar on day 0 stands out against the positive bars on the days before the announcement, visually capturing the reversal pattern.

4.2.2 Hypothesis Tests on Mean Cumulative Abnormal Returns

Table below reports the results of formal statistical tests on the mean CARs for each announcement type over both event windows. The test used is a one-sample t-test, which checks whether the average CAR across all events in a group is statistically different from zero. If the mean CAR is significantly different from zero, it means dividend announcements of that type are genuinely moving stock prices, not just generating random noise. Figure 4.4 shows the distribution of individual CARs across all events for each announcement type over the (-1, +1) window.

Table 4.4: Hypothesis Tests on Mean CAR by Announcement Type

Type	Window	N	Mean CAR (%)	SD (%)	SE (%)	t-stat	p-value	Sig.	CI Low (%)	CI High (%)
Decrease	(-1,+1)	244	-0.4085	8.4741	0.5425	-0.753	0.4522		-1.4771	0.6601

Increase	(-1,+1)	4,614	0.1316	5.1166	0.0753	1.747	0.0806	*	-0.0161	0.2793
Initiation	(-1,+1)	379	-0.1719	2.9881	0.1535	-1.120	0.2636		-0.4737	0.1299
Decrease	(-3,+3)	244	-0.6014	15.7648	1.0092	-0.596	0.5518		-2.5894	1.3866
Increase	(-3,+3)	4,615	0.2688	6.5428	0.0963	2.791	0.0053	***	0.0800	0.4577
Initiation	(-3,+3)	379	0.2896	8.6492	0.4443	0.652	0.5148		-0.5839	1.1632

*Note: One-sample t-test against H_0 : mean CAR = 0. CI = 95% confidence interval for the mean. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. SD = standard deviation of individual event CARs. SE = standard error of the mean CAR.*

H1 stated that dividend increases should generate positive abnormal returns around the announcement. The results provide support for this, particularly over the wider event window. Over the short (-1, +1) window, the mean CAR for increases is +0.13%, which is positive as expected, but only marginally significant at the 10% level ($t = 1.75$, $p = 0.081$). Over the broader (-3, +3) window, the evidence is much stronger the mean CAR rises to +0.27% and is significant at the 1% level ($t = 2.79$, $p = 0.005$). The 95% confidence interval over this window runs from +0.08% to +0.46%, meaning the entire range is above zero. This gives us reasonable confidence that dividend increases do generate positive abnormal returns, and H1 is supported, especially when the full seven-day window is considered.

H2 stated that dividend decreases should generate negative abnormal returns. The direction of the results is consistent with this prediction, the mean CAR is -0.41% over (-1, +1) and -0.60% over (-3, +3), both of which are negative as expected. However, neither result is statistically significant ($p = 0.452$ and $p = 0.552$ respectively).

The confidence intervals are very wide, for example ranging from -1.48% to +0.66% over the short window, which means we cannot rule out the possibility that the true mean is zero or even positive. The main reason for this is the small sample size of only 244 decrease events combined with extremely high variation across individual events the standard deviation of CARs for decreases is 8.47% over the short window, which is considerably higher than for the other groups. The direction of the result supports H2, but the evidence is not strong enough to formally confirm it. This is acknowledged as a limitation of the available data.

H3 stated that dividend initiations should generate positive abnormal returns. The results do not support this. Over the short window, the mean CAR is actually negative at -0.17%, and over the broader window it is positive at +0.29%, but neither is statistically significant ($p = 0.264$ and $p = 0.515$ respectively).

The confidence intervals include zero in both cases, meaning we cannot draw a firm conclusion either way. As discussed in Section 4.2.1, the initiation group has a complex pattern within the event window prices rise before the announcement and then fall sharply on the announcement day itself which may be why the cumulative return ends up close to zero over the short window and only modestly positive over the longer one. H3 is therefore not supported by the formal tests.

Figure 4.3: Distribution of CAR (-1, +1) by Announcement Type

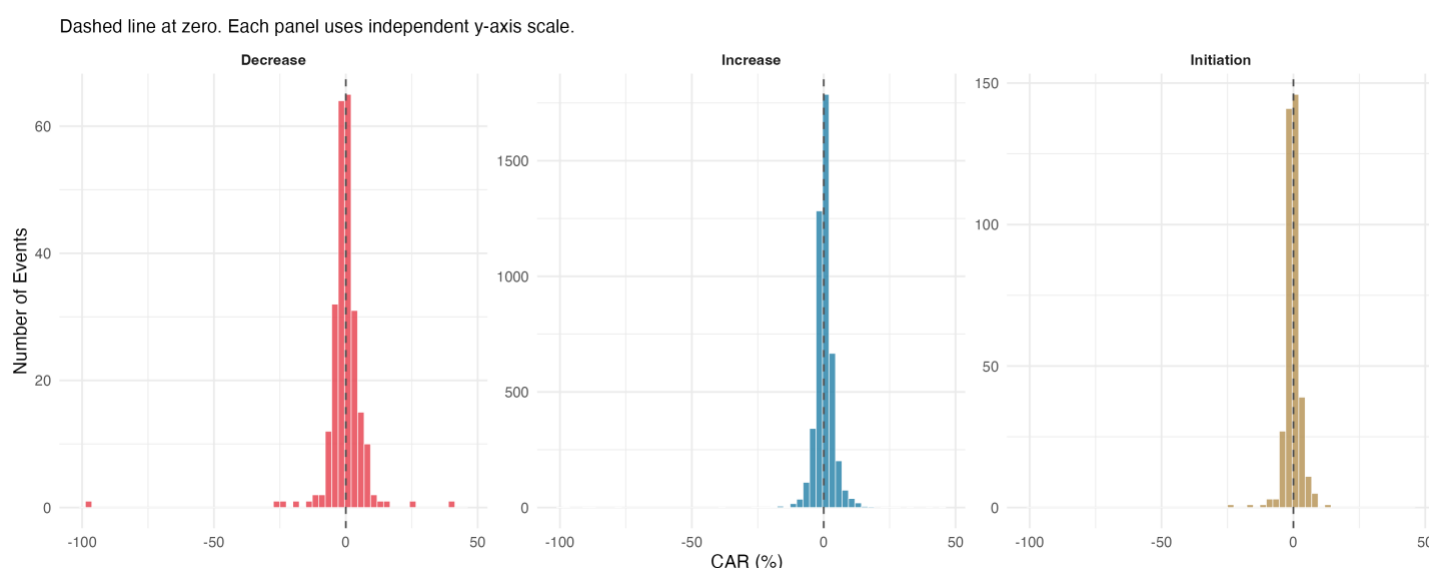


Figure 4.3 shows the spread of individual event CARs over the (-1, +1) window for each announcement type. The first thing to notice is that all three distributions are very tightly concentrated around zero, with most events generating CARs between -5% and +5%. This is consistent with what we found in the descriptive statistics in Section 4.1 while the average reactions are small, there is a great deal of variation at the individual event level.

The decrease panel on the left is the most spread out, with a small number of extreme outliers reaching as far as -100%, which explains why the mean for this group is negative but not statistically significant. The increase panel in the middle has the largest number of events and is the most tightly grouped, while the initiation panel on the right is similar in shape to the increase panel but based on far fewer observations.

4.2.3 Summary of Event Study Findings

To summarize the event study results, the findings show that not all dividend announcements generate the same market reaction. Dividend increases produce the clearest and most reliable positive market reaction, with the evidence being strongest over the $(-3, +3)$ window, which supports H1.

Dividend decreases move in the expected negative direction, but the small sample size and high variability within this group mean the result does not reach statistical significance, leaving H2 unconfirmed despite the directional consistency.

Dividend initiations produce no statistically significant reaction in either window, and the intra-window pattern of pre-announcement gains followed by a sharp reversal on day 0 makes the results difficult to interpret cleanly, meaning H3 is not supported.

These findings set the stage for Section 4.4, where a cross-sectional regression analysis examines whether firm-specific characteristics such as size, profitability, and valuation can help explain why the market reaction varies so much across individual dividend announcement events.

4.3 Diagnostic Tests

Following the event study results in Section 4.2, a set of diagnostic tests was performed to assess the statistical properties of the CAR distributions and the regression inputs. Three diagnostics are reported: normality of CARs (Section 4.3.1), multicollinearity among regression variables (Section 4.3.2), and the pairwise correlation structure of continuous control variables (Section 4.3.3). Together, these results justify specific methodological choices, most importantly the use of HC3 heteroskedasticity-robust standard errors in all regression models estimated in Section 4.4.

4.3.1 Normality of Cumulative Abnormal Returns

The normality of CAR was evaluated using two complementary tests: the Jarque-Bera (JB) test, which is applicable to samples of any size, and the Shapiro-Wilk (SW) test, which is reliable for samples up to 5,000 observations. The null hypothesis under both tests is that the underlying distribution is normal. The distributional shape of the CAR distributions for all groups and both event windows is presented visually in the Appendix. Results are reported in Table 4.5.

Table 4.5: Normality Tests on CAR Distributions

Type	Window	N	Skewness	Ex. Kurtosis	JB Statistic	JB p-value	JB Sig.	SW Statistic	SW p-value	SW Sig.
All Events	(-1,+1)	5,237	-8.8019	169.5546	6,340,843.75	<0.0001	***	-	-	N>5,000
Decrease	(-1,+1)	244	-5.9439	74.3052	57,569.61	<0.0001	***	0.5361	<0.0001	***
Increase	(-1,+1)	4,614	-9.0356	175.3365	5,973,103.91	<0.0001	***	0.5438	<0.0001	***
Initiation	(-1,+1)	379	-1.5521	15.2067	3,803.87	<0.0001	***	0.8468	<0.0001	***
All Events	(-3,+3)	5,238	-5.5089	179.3140	7,043,997.99	<0.0001	***	-	-	N>5,000
Decrease	(-3,+3)	244	-6.4957	77.3257	62,505.07	<0.0001	***	0.4229	<0.0001	***
Increase	(-3,+3)	4,615	-6.7493	130.1691	3,293,227.43	<0.0001	***	0.6186	<0.0001	***
Initiation	(-3,+3)	379	13.7199	237.5787	903,229.05	<0.0001	***	0.3246	<0.0001	***

*Note: H0: the distribution of CARs is normal. JB = Jarque-Bera test (all sample sizes). SW = Shapiro-Wilk test (N ≤ 5,000 only; shown as N>5,000 where not applicable). *** denotes rejection of H0 at the 1% significance level.*

The normality hypothesis is rejected at the 1% significance level for all groups and both event windows, with Jarque-Bera p-values of zero throughout. The Shapiro-Wilk statistic, reported for the Decrease, Increase, and Initiation subsamples, similarly yields p-values of zero in all cases. The magnitude of the JB statistics reaching into the millions for the larger subsamples reflects the extreme excess kurtosis documented in Table 4.2, consistent with the well-established finding in the event study literature that firm-level abnormal returns are typically leptokurtic.

4.3.2 Multicollinearity Among Regression Variables

To assess whether the regression specification in Section 4.4 is subject to problematic multicollinearity, Variance Inflation Factors (VIFs) were calculated for each regressor in the Model 2 specification.

Before presenting the results, it is useful to restate the full set of explanatory variables included in the model. The two primary variables of interest are dividend type dummies: Div_Increase, which takes a value of one for dividend increase announcements and zero otherwise, and Div_Initiation, which takes a value of one for dividend initiation announcements and zero otherwise.

Dividend decreases serve as the omitted reference category and are captured by the intercept. The three continuous control variables are FirmSize, measured as the natural logarithm of market capitalisation; ROA,

measured as net income divided by total assets; and PriceBook, measured as the market value of equity divided by the book value of equity.

Leverage was originally intended as a fourth control variable but was excluded due to insufficient data coverage, as discussed in Section 4.4. VIF values were calculated for all five retained regressors and are presented in Table 4.6.

Table 4.6: Variance Inflation Factors- Model 2 Specification

Window	Variable	VIF	Diagnosis
(-1,+1)	Div_Increase	2.4251	No concern
(-1,+1)	Div_Initiation	2.4115	No concern
(-1,+1)	FirmSize	1.0336	No concern
(-1,+1)	ROA	1.0447	No concern
(-1,+1)	PriceBook	1.0419	No concern
(-3,+3)	Div_Increase	2.4251	No concern
(-3,+3)	Div_Initiation	2.4115	No concern
(-3,+3)	FirmSize	1.0336	No concern
(-3,+3)	ROA	1.0447	No concern
(-3,+3)	PriceBook	1.0419	No concern

Note: Model specification: $CAR \sim Div_Increase + Div_Initiation + FirmSize + ROA + PriceBook$. Reference category: Dividend Decrease. VIF thresholds: <2.5 = No concern; $2.5-5.0$ = Mild; $5.0-10.0$ = Moderate; >10.0 = Severe. Results are identical across both event windows as the regressor matrix is the same.

All VIF values are below 2.5 across both event windows, indicating no multicollinearity concern in the regression model. The dividend type dummies (Div_Increase: 2.43; Div_Initiation: 2.41) show the highest VIFs, which is expected given that together they jointly represent the three mutually exclusive announcement categories.

The continuous control variables, FirmSize (1.03), ROA (1.04), and PriceBook (1.04) exhibit VIFs very close to unity, indicating that they are Unrelated to one another and to the dummy variables. These findings confirm that the regression estimates reported in Section 4.4 are not materially distorted by collinearity among the regressors.

4.3.3 Pairwise Correlation of Continuous Control Variables

As a supplementary check, Table 4.7 reports the pairwise Pearson correlation coefficients among the three continuous control variables: the natural logarithm of market capitalisation (FirmSize), return on assets (ROA), and the price-to-book ratio (PriceBook).

Table 4.7: Pairwise Pearson Correlations Among Continuous Control Variables

Variable	FirmSize	ROA	PriceBook
FirmSize	1.0000	0.0492	0.0917
ROA	0.0492	1.0000	0.1803
PriceBook	0.0917	0.1803	1.0000

Note: Computed on the regression subsample for the (-1, +1) event window. FirmSize = $\ln(\text{Market Capitalisation})$. ROA = Net Income / Total Assets. PriceBook = Market Value of Equity / Book Value of Equity.

All pairwise correlations are low, with the highest being between ROA and PriceBook ($r = 0.18$), consistent with the well-known positive relationship between profitability and market-to-book ratios (Fama and French, 1993). The correlations between FirmSize and ROA ($r = 0.05$) and between FirmSize and PriceBook ($r = 0.09$) are negligible. These results confirm the VIF findings and provide additional assurance that the control variables capture distinct dimensions of firm-level heterogeneity without introducing redundancy into the regression.

Taken together, the diagnostic tests in Sections 4.3.1 to 4.3.3 establish that:

- (i) CAR distributions are non-normal across all subgroups, motivating robust standard errors;
- (ii) the regression specification is free of multicollinearity; and
- (iii) the continuous control variables are largely uncorrelated, supporting their joint inclusion in the model. The analysis now proceeds to the cross-sectional regression results in Section 4.4.

4.4 Cross-Sectional Regression Results

The event study in Section 4.2 told us whether dividend announcements moved stock prices on average. This section now asks a different question: among the thousands of individual announcement events, why do some generate larger price reactions than others?

To answer this, a cross-sectional regression is run with the cumulative abnormal return (CAR) of each individual event as the dependent variable. The regression tests whether the type of dividend announcement and a set of firm characteristics like company size, profitability, and price-to-book ratio can explain the variation in CARs across events.

The non-normality of CAR distributions documented in Section 4.3.1 has direct implications for the cross-sectional regression analysis. For the dividend increase subsample ($N = 4,614$), the Central Limit Theorem ensures that the sampling distribution of the mean converges to normality, and standard inference remains valid. However, for the dividend decrease ($N = 244$) and initiation ($N = 379$) subsamples, the combination of smaller sample sizes and pronounced non-normality raises concerns about the reliability of conventional standard errors. Furthermore, the residual diagnostic plots presented in Section 4.4.4 confirm the presence of heteroskedasticity in the regression residuals. For these reasons, all models are estimated using HC3 heteroskedasticity-consistent robust standard errors, which remain reliable under non-normality and unequal error variances without requiring any distributional assumptions about the residuals.

Three models are estimated for each event window.

Model 1 includes only the dividend type dummy variables, with dividend decreases as the reference category. This means the intercept in every model represents the average CAR for dividend decrease events, and the coefficients on Dividend Increase and Dividend Initiation show how much higher or lower the CARs for those types are compared to decreases. Model 2 adds the three firm-level control variables.

Model 3 further adds industry fixed effects to account for any systematic differences across sectors. It is important to note that leverage (debt-to-equity ratio) was originally intended as a fourth control variable, but it was excluded from all models due to a large proportion of missing values in the dataset. It would have resulted in a significant loss of observations, reducing the reliability of the regression. This omission is acknowledged as a data limitation and is discussed further in future chapters.

All models use HC3 robust standard errors, as motivated by the non-normality and heteroskedasticity identified in the diagnostic tests in Section 4.3. Coefficients are reported as percentages for ease of interpretation. T-statistics are shown in parentheses below each coefficient.

4.4.1 Panel A Event Window (-1, +1)

Table below presents the regression results for the short event window (-1, +1).

Table 4.8: Cross-Sectional Regression Results Panel A, Event Window (-1, +1)

Variable	Model 1	Model 2	Model 3
(Intercept) Div. Decrease baseline	-0.437	-0.973	-1.216
	<i>(-0.778)</i>	<i>(-0.800)</i>	<i>(-0.817)</i>
Div_Increase	0.588	0.536	0.543
	<i>(1.039)</i>	<i>(0.971)</i>	<i>(0.989)</i>
Div_Initiation	0.304	0.285	0.300
	<i>(0.523)</i>	<i>(0.485)</i>	<i>(0.513)</i>
FirmSize	-	0.118	0.144
		<i>(0.596)</i>	<i>(0.715)</i>
ROA	-	1.059	1.079
		<i>(1.079)</i>	<i>(1.059)</i>
PriceBook	-	-0.001	-0.001
		<i>(-0.619)</i>	<i>(-0.679)</i>
Industry Fixed Effects	No	No	Yes
N	5,055	5,055	5,055
R ²	0.0008	0.0012	0.0015
Adj. R ²	0.0004	0.0002	-0.0006

Note: Dependent variable is the individual event CAR (%), expressed as a percentage. Coefficients represent percentage point differences relative to dividend decreases (reference category). T-statistics are in parentheses (in italics). Robust SE = HC3 heteroskedasticity-consistent standard errors. No coefficients achieve significance at the 10% level. Leverage (Debt-to-Equity) was excluded due to insufficient data coverage.- = variable not included in that model.

Looking at the intercept first which represents the baseline average CAR for dividend decrease events, it is negative across all three models, ranging from -0.437% in Model 1 to -1.216% in Model 3. This is directionally consistent with H2, which predicted that decreases would generate negative abnormal returns. However, none

of these intercepts are statistically significant, which is consistent with what was found in the event study hypothesis tests in Section 4.2.2.

The coefficient on Div_Increase is positive in all three models, sitting between 0.536% and 0.588% across Models 1 to 3. This tells us that, after controlling for firm size, profitability, and valuation, events involving dividend increases tend to generate CARs that are roughly 0.5 percentage points higher than events involving dividend decreases. The direction is consistent with H1. However, none of these coefficients reach statistical significance the t-statistics are all around 1.0 and the p-values are well above 0.10. Similarly, the Div_Initiation coefficient is positive across all models (0.285% to 0.304%) but is even further from significance, with t-statistics below 0.53.

Turning to the control variables in Model 2, none of them are statistically significant either. FirmSize has a small positive coefficient of 0.118% ($t = 0.60$), which actually runs against the theoretical expectation that larger firms should generate smaller reactions due to lower information asymmetry. ROA shows a positive coefficient of 1.059% ($t = 1.08$), which is in the expected direction, more profitable firms may have more credible dividend signals but does not reach significance. PriceBook has a near-zero coefficient of -0.001% ($t = -0.62$), which is essentially flat. Adding industry fixed effects in Model 3 barely changes any of the coefficients, and the adjusted R^2 actually turns negative (-0.0006), indicating that the industry dummies are adding noise rather than improving the model.

The overall fit of all three models is poor. The R^2 values range from just 0.08% to 0.15%, meaning that the variables included in the regression explain less than 0.2% of the variation in individual event CARs.

The absence of statistically significant coefficients across all three models is attributable to several factors. First, individual event CARs carry a standard deviation of over 5% around a mean of less than 0.2%, producing a very low signal-to-noise ratio that makes it difficult to detect cross-sectional patterns at the coefficient level.

Second, the S&P 500 sample is inherently homogeneous: firm size, profitability, and valuation vary far less within a large-cap index than they would in a broader market sample, limiting the explanatory power of these variables.

Third, the magnitude of any individual announcement's market reaction is driven primarily by event-specific factors not captured in the model, such as the degree of surprise relative to analyst expectations, concurrent news releases, and prevailing market conditions at the time of the announcement.

Finally, the exclusion of leverage due to insufficient data coverage removes one theoretically motivated cross-sectional driver from the specification entirely. Taken together, these factors explain why the regression identifies the correct directional relationships but cannot achieve statistical significance at the individual coefficient level.

The variation in CARs across individual announcements is heavily driven by firm-specific and event-specific factors that are not captured by broad firm characteristics such as size or profitability. The finding suggests that the aggregate patterns identified in the event study particularly the positive reaction to dividend increases are a consistent feature of the full sample, but the magnitude of the reaction in any given announcement event is difficult to predict from observable firm-level data alone.

4.4.2 Panel B Event Window (-3, +3)

Table below presents the same three models estimated over the wider (-3, +3) event window.

Table 4.9: Cross-Sectional Regression Results- Panel B, Event Window (-3, +3)

Variable	Model 1	Model 2	Model 3
(Intercept)-Div. Decrease baseline	-0.763	-0.631	-0.829
	<i>(-0.726)</i>	<i>(-0.338)</i>	<i>(-0.381)</i>
Div_Increase	1.043	1.019	1.051
	<i>(0.989)</i>	<i>(0.982)</i>	<i>(1.009)</i>
Div_Initiation	1.128	1.097	1.158
	<i>(0.986)</i>	<i>(0.948)</i>	<i>(0.988)</i>
FirmSize	-	-0.038	0.006
		<i>(-0.139)</i>	<i>(0.022)</i>
ROA	-	0.844	1.249
		<i>(0.578)</i>	<i>(0.835)</i>

PriceBook	-	0.000	0.000
		(0.058)	(0.169)
Industry Fixed Effects	No	No	Yes
N	5,056	5,056	5,056
R ²	0.0009	0.0010	0.0017
Adj. R ²	0.0005	0.0002	-0.0005

Note: Dependent variable is the individual event CAR (%), expressed as a percentage. Coefficients represent percentage point differences relative to dividend decreases (reference category). T-statistics are in parentheses (in italics). Robust SE = HC3 heteroskedasticity-consistent standard errors. No coefficients achieve significance at the 10% level. Leverage (Debt-to-Equity) was excluded due to insufficient data coverage. - = variable not included in that model.

The results over the wider window follow a very similar pattern to Panel A, but with larger coefficient estimates. The intercept representing the average CAR for dividend decreases is again negative across all models, ranging from -0.631% to -0.829%, still without statistical significance. The Div_Increase coefficient is larger here, sitting between 1.019% and 1.051% across the three models, suggesting that over the full seven-day window increases generate about a percentage point higher CARs than decreases on average. The Div_Initiation coefficient is similarly larger, ranging from 1.097% to 1.158%. Despite these larger magnitudes, none of the coefficients achieve significance, all t-statistics remain close to 1.0 and all p-values are well above 0.10.

The firm control variables in Model 2 are again all insignificant. FirmSize switches sign across the two windows positive in Panel A and slightly negative in Panel B (-0.038%, $t = -0.14$) which suggests it has no consistent relationship with CAR magnitude in either direction. ROA and PriceBook remain close to zero and insignificant. The R² values are marginally higher than in Panel A (0.09% to 0.17%) but are still essentially zero, confirming that the model has no meaningful explanatory power. The results across both panels are therefore broadly consistent, and the conclusions drawn from the short window analysis hold equally for the wider window.

4.4.3 Industry Fixed Effects

The table below reports the individual industry fixed effect coefficients from Model 3 for both event windows. The reference category is the Communication Services sector, so each coefficient shows the average difference in CAR between that sector and Communication Services firms, after controlling for dividend type and firm characteristics

Table 4.10: Industry Fixed Effects Coefficients Model 3

Industry (ref: Communication Services)	(-1,+1) Coef (%)	(-1,+1) t-stat	(-3,+3) Coef (%)	(-3,+3) t-stat
Consumer	-0.081	-0.127	-0.152	-0.198
Energy	0.018	0.027	-0.206	-0.257
Financials	0.157	0.261	0.163	0.221
Healthcare	-0.071	-0.116	-0.410	-0.518
Industrials	0.156	0.251	-0.199	-0.266
Technology	0.197	0.312	0.091	0.120

Note: Reference category = Communication Services. Coefficients expressed as percentages. All coefficients are statistically insignificant at all conventional levels ($p > 0.10$ throughout). Results are from Model 3 for both event windows.

None of the industry fixed effects are statistically significant in either window. The coefficients are small and inconsistent in sign for example, Energy switches from a small positive 0.018% over $(-1, +1)$ to a negative -0.206% over $(-3, +3)$. Financials shows positive coefficients in both windows (0.157% and 0.163%), while Healthcare is consistently negative (-0.071% and -0.410%), but again neither achieves significance.

The absence of any significant industry effects suggests that the market reaction to dividend announcements among S&P 500 firms does not systematically differ across sectors, at least after controlling for announcement type and basic firm characteristics. This is a reasonable finding given that the sample is restricted to large-cap firms within a single index, which tend to be relatively homogeneous in terms of investor base, analyst coverage, and disclosure standards compared to a broader cross-section of listed companies.

4.4.4 Regression Diagnostics

Figures 4.4 and 4.5 present the residual diagnostic plots for Model 2 estimated over the $(-1, +1)$ window, which is the main specification of interest. These plots were examined to assess whether the regression assumptions are sufficiently satisfied and to confirm that HC3 robust standard errors were the appropriate choice.

Figure 4.4: Residuals vs Fitted Values Model 2, Window (-1, +1)

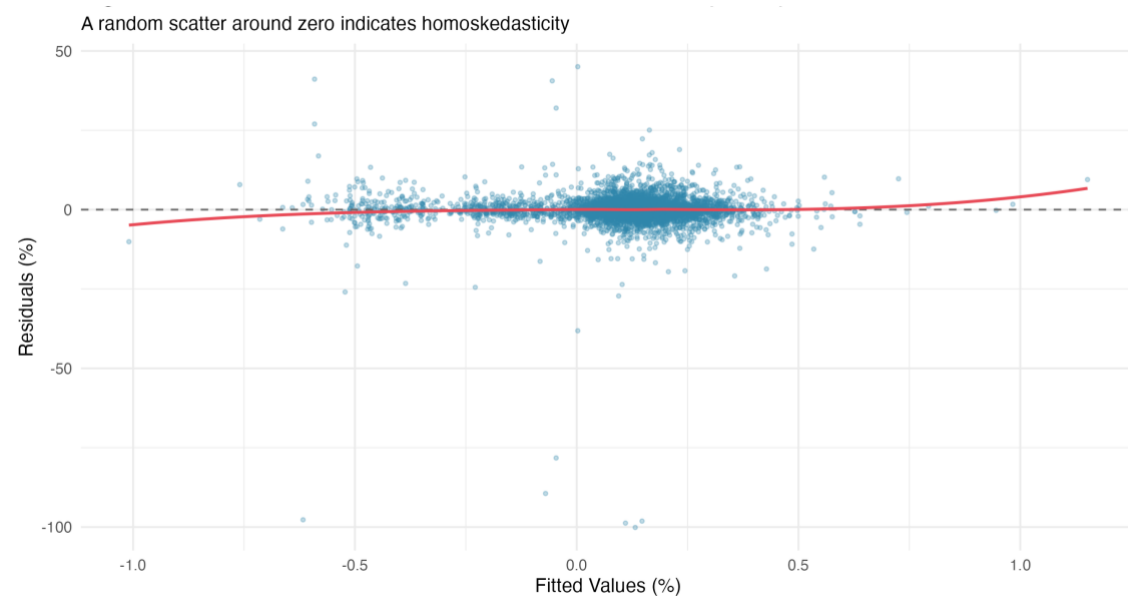


Figure 4.4 plots the regression residuals against the fitted values from Model 2. In an ideal regression, these points would be randomly scattered around zero with no clear pattern. The plot here shows that the vast majority of observations are clustered very tightly around a fitted value of 0%, which reflects the fact that the model predicts very little variation in CARs essentially predicting close to zero for almost every event.

The spread of residuals is noticeably wider on the left side of the chart (negative fitted values) and more compressed in the middle, suggesting some degree of heteroskedasticity that is, the variability of the residuals is not constant across all fitted values.

A small number of extreme outliers are also visible, including one observation with a residual below -100%. The red trend line runs very close to zero throughout but with a very slight upward slope, which is consistent with the near-zero but positive coefficients on the dividend type variables. Overall, this plot confirms that heteroskedasticity is present, which validates the decision to use HC3 robust standard errors throughout the analysis.

Figure 4.5: Q-Q Plot of Regression Residuals Model 2, Window (-1, +1)

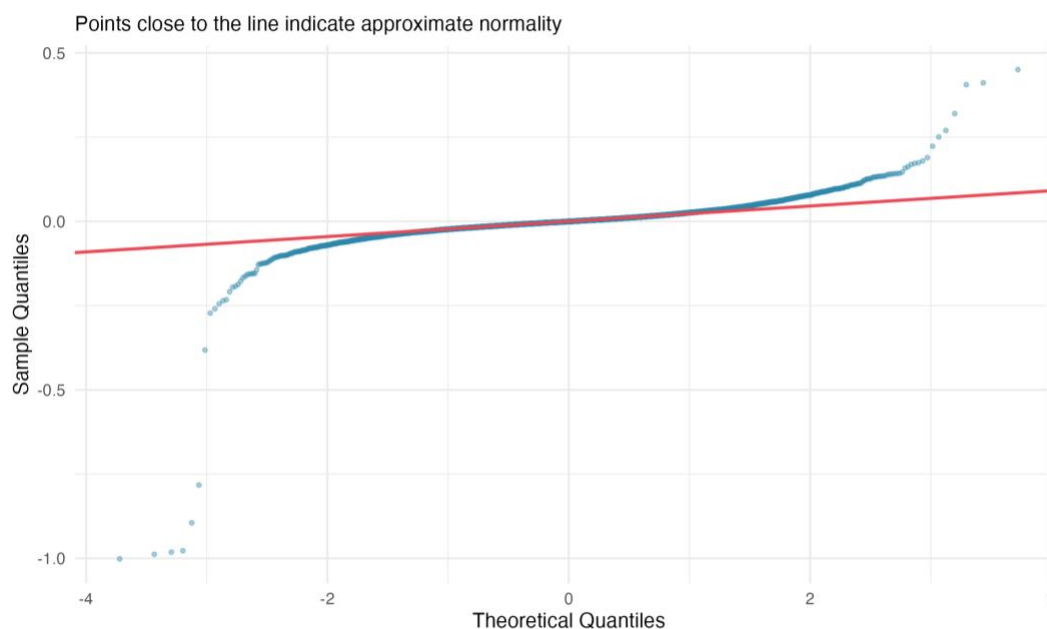


Figure 4.5 is a Q-Q plot, which compares the distribution of the actual regression residuals against what would be expected if the residuals were perfectly normally distributed. If normality held, all points would fall along the red diagonal line. In the middle of the chart which covers the majority of observations, the points track the line reasonably closely, indicating that most residuals behave approximately normally.

However, at both ends of the chart the points deviate substantially from the line, with the left tail dropping far below and the right tail rising sharply above it. This is the classic shape of a fat-tailed distribution, consistent with what was already found in the normality tests in Section 4.3.1.

The deviation is most pronounced at the extremes, confirming that a small number of events with very large positive or negative CARs are pulling the tails away from normality, this reinforce the importance of using robust standard errors, which are designed to remain reliable even when the residuals are not perfectly normally distributed.

4.4.5 Summary of Regression Findings

Although none of the regression coefficients achieve statistical significance, the analysis is not without value. The cross-sectional regression addresses a fundamentally different question from the event study not whether dividend announcements move stock prices on average, but whether observable firm characteristics can explain why some announcements generate larger reactions than others.

The answer from this analysis is clearly no, and that is itself a substantive finding. It suggests that among S&P 500 firms, the magnitude of the market reaction to a dividend announcement cannot be reliably predicted from annual measures of firm size, profitability, or valuation. For investors, this implies that no systematic trading strategy based on these characteristics is likely to exploit dividend announcement returns. For researchers, it points toward event-specific factors such as the degree of surprise relative to analyst forecasts or the presence of concurrent news as more promising explanatory variables than firm-level accounting measures.

The regression therefore complements the event study by defining the boundaries of what firm characteristics can and cannot explain about dividend announcement reactions in this market.

The cross-sectional regression results can also be summarized in three main points. First, the direction of the coefficients is broadly consistent with the predictions of dividend signaling theory and with the event study findings in Section 4.2 dividend increases and initiations are associated with higher CARs relative to decreases across all models and both event windows.

Second, none of the coefficients are statistically significant at any conventional level, meaning we cannot confidently attribute the variation in individual event CARs to the firm characteristics and announcement types included in the model.

Third, the overall explanatory power of the models is negligible with R^2 values below 0.2% which indicates that the variation in individual CARs is driven primarily by idiosyncratic factors that are not captured by firm size, profitability, valuation, or industry sector. These findings are consistent with a large body of event study literature that finds dividend announcement effects to be real and significant at the aggregate level, while remaining difficult to explain cross-sectionally using standard firm-level characteristics.

4.5 Chapter Summary

This chapter presented the full set of empirical results from the study. The findings are drawn from three sources: descriptive statistics, the event study analysis, and the cross-sectional regression. This section brings all of those results together in one place.

Table 4.11 below provides a concise summary of how the evidence lines up against each of the three hypotheses across both the event study and the regression analysis.

Table 4.11: Summary of Hypothesis Test Results

Hypothesis	Prediction	Event Study (-1,+1)	Event Study (-3,+3)	Regression (Cross-Sectional)	Verdict
H1: Dividend Increases	Positive abnormal returns	Mean CAR = +0.13% t = 1.75, p = 0.081 (10% level only)	Mean CAR = +0.27% t = 2.79, p = 0.005 (1% level)	Coef. positive (~0.54%) but not significant (p > 0.10)	Supported (especially over -3,+3 window)
H2: Dividend Decreases	Negative abnormal returns	Mean CAR =- 0.41% t =-0.75, p = 0.452 (not significant)	Mean CAR =- 0.60% t =-0.60, p = 0.552 (not significant)	Intercept negative but not significant (p > 0.10)	Not confirmed (correct direction, small sample)
H3: Dividend Initiations	Positive abnormal returns	Mean CAR =- 0.17% t =-1.12, p = 0.264 (not significant)	Mean CAR = +0.29% t = 0.65, p = 0.515 (not significant)	Coef. positive (~1.10%) but not significant (p > 0.10)	Not supported (mixed pattern across windows)

The clearest finding in this study is that dividend increases generate a genuine and statistically significant positive market reaction. The evidence is strongest over the seven-day window, where the mean cumulative abnormal return of +0.27% is significant at the 1% level. The reaction is also well-timed; prices stay flat before the announcement and rise sharply the day after, which is consistent with the market processing new information quickly and efficiently.

For dividend decreases, the results move in the expected negative direction, but statistical significance is not achieved in either window. This is largely a sample size problem with only 244 decrease events and very high

variation across individual cases, the data does not provide enough precision to confirm the result formally. The direction of the finding is still meaningful and consistent with theory.

Dividend initiations produced the most complex pattern. Prices rose in the days before the announcement but then fell sharply on the announcement day itself, resulting in a mixed cumulative return depending on which window is used. Neither window produced a statistically significant result, and H3 is therefore not supported. The pre-announcement build-up and same-day reversal is an interesting finding in its own right and is discussed further in the coming chapters.

The cross-sectional regression added useful context. While the coefficients consistently pointed in the right direction increases and initiations associated with higher CARs than decreases, none were statistically significant and the overall model fit was extremely low (R^2 below 0.2%). This tells us that while dividend type matters at the aggregate level, the size of the market reaction in any single announcement event is driven by factors that standard firm characteristics cannot easily capture.

Taken together, the results provide meaningful evidence that dividend announcements carry information that the stock market responds to, particularly for dividend increases. The findings align broadly with dividend signaling theory and suggest that, among S&P 500 firms over the 2011 to 2025 period, the market does distinguish between different types of dividend news. Chapter 5 now turns to interpreting what these results mean in the wider context of existing theory and literature.

CHAPTER 5

Discussion

This chapter interprets the empirical findings presented in Chapter 4 in light of the theoretical frameworks and prior literature reviewed in Chapters 2 and 3. The event study results provide evidence that dividend announcements made by S&P 500 firms generate measurable stock market reactions, though the strength and statistical reliability of these reactions differ substantially across announcement types. The cross-sectional regression analysis adds a further layer of insight by examining whether firm-level characteristics can explain the

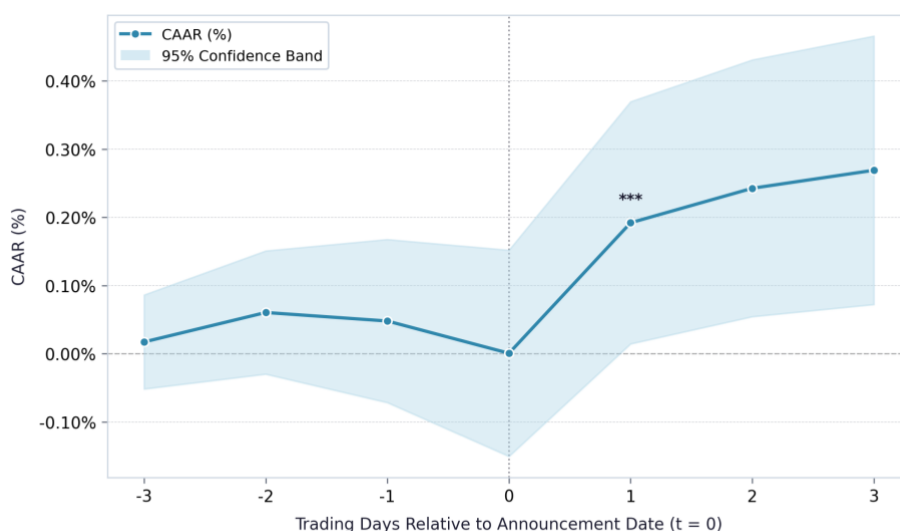
variation in market reactions across individual announcement events. The discussion is organized around the three hypotheses before addressing the broader implications for market efficiency and the relationship between the findings and existing evidence in the dividend announcement literature.

5.1 Market Reaction to Dividend Increase

The results provide support for H1, which predicted that dividend increases would generate positive abnormal stock returns around the announcement date. Over the short event window $(-1, +1)$, the mean cumulative abnormal return is $+0.13\%$, significant at the 10% level ($t = 1.75$, $p = 0.081$). The evidence is considerably stronger over the wider $(-3, +3)$ window, where the mean CAAR rises to $+0.27\%$ and is significant at the 1% level ($t = 2.79$, $p = 0.005$), with a 95% confidence interval that lies entirely above zero. These results are consistent with the predictions of dividend signaling theory, which holds that dividend increases represent credible signals of management confidence in future earnings sustainability (Bhattacharya, 1979; Miller and Rock, 1985).

Figure 5.1 plots the CAAR for dividend increases across the seven-day event window alongside the 95% confidence band. The figure makes three things immediately visible. First, the CAAR line remains essentially flat and the confidence band stays close to zero throughout the pre-announcement period, confirming that no systematic price movement occurs before the announcement date. Second, the CAAR remains at zero on the announcement day itself before jumping sharply on day $+1$, where the AAR is the strongest and most precisely estimated result in the entire analysis, marked by the *** significance indicator confirming significance at the 1% level. Third, following day $+1$ the CAAR continues to rise gradually through days $+2$ and $+3$, but the confidence band widens considerably over this period, reflecting greater uncertainty in the cumulative return estimate beyond the initial reaction.

Figure 5.1: CAAR Around Dividend Increase Announcement



Note: Significance stars reflect daily AAR t-tests. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. $N = 4,614$ events. Market model estimated over $[-250, -30]$ trading days.

*Note: Significance stars reflect daily AAR t-tests. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. $N = 4,614$ dividend increase events. Market model estimated over $[-250, -30]$ trading days. Shaded area represents the 95% confidence band based on cumulative standard errors of daily AAR estimates. S&P 500 index used as market proxy.*

The daily abnormal return pattern provides additional insight into the mechanics of the market reaction. The pre-announcement CAAR for increases remains essentially flat through to day -1, with no statistically significant daily AARs in the three days prior to the announcement. This absence of pre-announcement drift is consistent with semi-strong form market efficiency and suggests that dividend increase decisions are not systematically anticipated by investors ahead of formal disclosure.

The market reaction is concentrated almost entirely on day +1, where the AAR is the strongest and most precisely estimated result in the entire analysis. The reaction appears on day +1 rather than day 0 first because announcements made after market close cannot be priced until the following morning, deferring the reaction to day +1. Also the slow incorporation of information, where some investors require additional time to process and act on new disclosures.

The pattern observed here is more consistent with the first explanation. Under slow incorporation one would expect a gradual price adjustment spread across several days, whereas the data shows a sharp concentration on day +1 followed by immediate stabilisation with no further drift through day +3. This is consistent with the evidence in Aharony and Swary (1980), who document using daily data that abnormal returns around dividend announcements are rapid and concentrated rather than gradual.

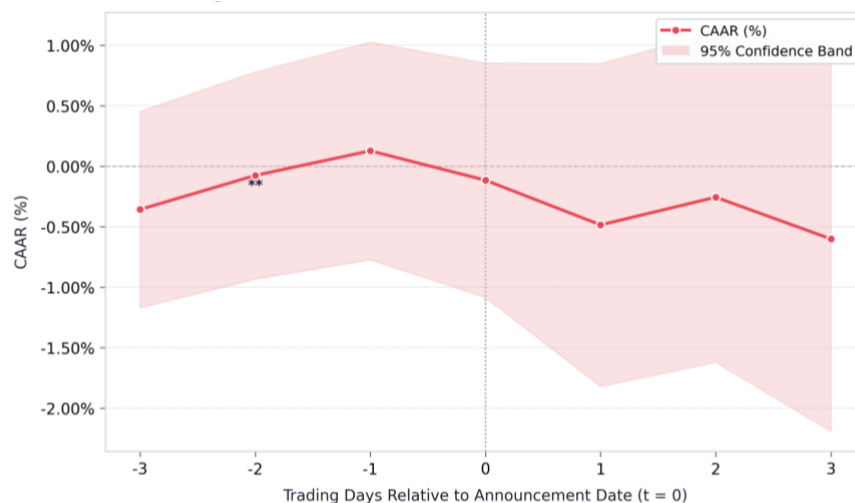
Following day +1, the CAAR stabilizes and shows no further drift through day +3, which is precisely what semi-strong form market efficiency predicts: prices incorporate new information rapidly and completely, leaving no exploitable pattern thereafter.

5.2 Market Reaction to Dividend Decreases

The results for dividend decreases are directionally consistent with H2 but do not achieve statistical significance in either event window. The mean CAR is negative at both -0.41% over $(-1, +1)$ and -0.60% over $(-3, +3)$, consistent with the theoretical prediction that decreases convey unfavorable signals about a firm's financial condition. However, with p-values of 0.452 and 0.552 respectively, neither result can be formally confirmed.

Figure 5.2 below makes the statistical challenge immediately visible. The confidence band is very wide throughout the entire event window and stays around the zero line at every point, meaning the cumulative return cannot be distinguished from zero with any reasonable degree of confidence at any day in the window. This is in sharp contrast to the narrow and decisive band observed for dividend increases in Figure 5.1, and reflects the different nature of the decrease subsample.

Figure 5.2: CAAR Around Dividend Decrease Announcement



Note: Significance stars reflect daily AAR t-tests. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. $N = 244$ events. Market model estimated over $[-250, -30]$ trading days. Shaded area = 95% confidence band.

Note: Significance stars reflect daily AAR t-tests. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. $N = 244$ dividend decrease events. Market model estimated over $[-250, -30]$ trading days. Shaded area represents the 95% confidence band based on cumulative standard errors of daily AAR estimates. S&P 500 index used as market proxy.

Two factors explain this outcome. First, the decrease subsample is small, with only 244 retained events compared to 4,614 for increases. With a standard deviation of CARs exceeding 8% in this group, the sample lacks the statistical power to detect moderate-sized effects.

Second, dividend decreases among S&P 500 firms are not a homogeneous category. Some cuts reflect acute financial distress and are interpreted by the market as a negative signal about future earnings, consistent with signaling theory. Others represent deliberate strategic decisions such as redirecting cash toward share repurchases, acquisitions or capital investment programmes that investors may interpret as a sign of growth confidence rather than financial weakness. Evidence shows that firms do reduce dividends strategically to fund repurchases rather than as a signal of financial difficulty, meaning not all cuts carry the same negative informational content (Grullon and Michaely, 2002).

Among large S&P 500 firms, which have the financial sophistication and strategic optionality to make such choices, this distinction is particularly relevant. The market appears to price each cut differently depending on the perceived reason behind it, which is why some individual decrease events generate large negative CARs while others generate neutral or even positive reactions. This heterogeneity inflates the variance of individual event CARs and dilutes the average reaction.

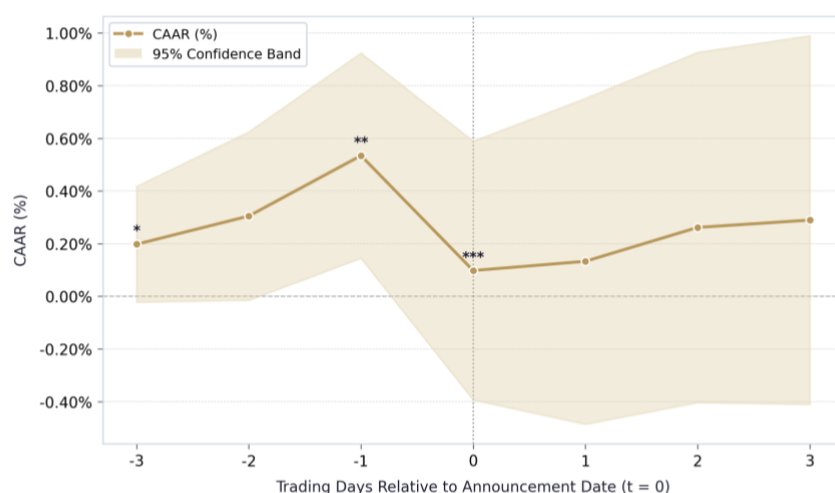
While the direction of the result is meaningful and consistent with signaling theory, the formal hypothesis must remain unconfirmed given the data constraints.

5.3 Market Reaction to Dividend Initiations

Dividend initiations produce the most complex and difficult to interpret results in the study. H3 predicted positive abnormal returns around initiation announcements, consistent with the life-cycle hypothesis that initiations signal a structural transition toward financial maturity and stable cash-flow generation (Grullon, Michaely and Swaminathan, 2002). The results do not support this prediction statistically, with mean CARs of -0.17% over (-1, +1) and +0.29% over (-3, +3), neither of which approaches statistical significance ($p = 0.264$ and $p = 0.515$ respectively).

Figure 5.3 below captures the most visually distinctive pattern in the entire study. Rather than the clean single-direction reaction seen for increases, the CAAR for initiations traces a rise-and-fall shape that makes the event window choice critical for interpreting the result.

Figure 5.3: CAAR Around Dividend Initiation Announcement



Note: Significance stars reflect daily AAR t-tests. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. $N = 379$ events. Market model estimated over $[-250, -30]$ trading days. Shaded area = 95% confidence band.

Note: Significance stars reflect daily AAR t-tests. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. $N = 379$ dividend initiation events. Market model estimated over $[-250, -30]$ trading days. Shaded area represents the 95% confidence band based on cumulative standard errors of daily AAR estimates. S&P 500 index used as market proxy.

In the days before the announcement, the CAAR climbs steadily to 0.53% by day -1, supported by marginally significant positive AARs on day -3 ($p = 0.079$) and day -1 ($p = 0.045$). This pre-announcement build-up suggests that some market participants may anticipate the initiation ahead of its formal disclosure, possibly

through observable changes in the firm's financial position such as improving earnings stability or declining investment intensity. However, on the announcement day itself, the CAAR reverses sharply as the AAR drops to -0.44%, which is the largest single-day negative reaction in the entire analysis and is significant at the 1% level ($t = -2.86$, $p = 0.005$).

Two interpretations are possible. The first is a sell-the-news effect, where investors who accumulated positions in anticipation of the announcement liquidate upon confirmation of the news. The second is that the actual initiation announcement failed to meet the elevated expectations already priced in during the pre-announcement period.

From an economic perspective, dividend initiations are inherently more complex signals than increases or decreases. Under the life-cycle hypothesis of Grullon, Michaely and Swaminathan (2002), a firm that initiates a dividend is signaling that it has reached a stage of financial maturity where it no longer has sufficient high-return investment opportunities to deploy all of its internally generated cash. While this transition from a growth firm to a mature dividend-paying firm can be viewed positively as a sign of financial stability, it can also be interpreted less favorably by growth-oriented investors who may prefer the firm to reinvest its earnings rather than distribute them.

This mixed signal means that the informational content of an initiation announcement is less clear-cut than that of a dividend increase, and different investors may draw very different conclusions from the same announcement depending on their view of the firm's growth prospects. Among large S&P 500 firms, which are already well-established and heavily covered by analysts, the initiation signal may carry even less surprise value than it would for smaller less well-known companies, further contributing to the mixed and inconclusive market reaction observed here.

The CAAR partially recovers through days +1 to +3, ultimately settling at +0.29% over the full window, but the complexity of the intra-window pattern means the choice of event window materially affects the sign of the cumulative return, making the result difficult to interpret as either a clear positive or negative market reaction.

5.4 Cross-Sectional Regression Results

The cross-sectional regression analysis finds that neither announcement type dummies nor firm-level control variables produce statistically significant coefficients in any of the three model specifications or across either event window. The R-squared values remain below 0.2% throughout, indicating negligible explanatory power at the individual event level.

This result addresses a fundamentally different question from the event study. Rather than asking whether dividend announcements move prices on average, the regression asks whether observable firm characteristics can explain why some announcements generate larger reactions than others. The answer is clearly no, and understanding why requires going beyond the numbers.

The most immediate explanation is statistical. Individual event CARs carry a standard deviation exceeding 5% against a mean of less than 0.3%, producing a very low signal-to-noise ratio. When individual returns vary this widely relative to the average, a regression model built on annual accounting data is unlikely to explain a meaningful share of that variation regardless of which variables are included.

The deeper explanation is economic. S&P 500 firms are among the most closely followed companies in the world, covered by many analysts and scrutinized by institutional investors on a daily basis. Because so much information about these firms is already publicly available, the gap between what managers know and what the market already knows is naturally smaller than it would be for less visible companies. When the informational environment is already relatively uniform across all firms in the sample, differences in size, profitability or valuation have little room to explain differences in market reactions because the signaling value of any given announcement is already compressed across the board.

This interpretation is supported by Amihud and Li (2006), who document that the price reaction to dividend news is smaller in firms with high institutional holdings because institutional investors are better informed and trade ahead of announcements, reducing the surprise component that drives the reaction. Since S&P 500 firms are among the most heavily institutionally owned in the market, the variation in individual event CARs within this sample is driven primarily by announcement-specific and timing factors rather than by observable firm characteristics. This is precisely what the regression results confirm.

For investors, the implication is that no trading strategy based on firm size, profitability or valuation is likely to generate reliable returns around dividend announcements within the S&P 500. For future research, incorporating analyst forecast data directly into the cross-sectional specification would allow announcement-specific expectations to be measured explicitly, which represents the most logical and promising extension of this work.

5.5 Implications for Market Efficiency

The timing and pattern of the observed market reactions provide us with useful evidence on the informational efficiency of the S&P 500. The results for dividend increases are particularly informative in this regard. The absence of pre-announcement drift in the CAAR for increases, combined with a sharp and concentrated reaction on day +1 followed by immediate price stabilization, is consistent with the semi-strong form of the Efficient Market Hypothesis as formulated by Fama (1970). Under semi-strong efficiency, stock prices should incorporate publicly released information rapidly and fully, leaving no systematic post-announcement drift. The pattern observed for increases matches this prediction precisely.

The results for initiations are more difficult to reconcile with strict market efficiency. The pre-announcement build-up in CAAR over days -3 to -1 raises the possibility that some participants trade on information about forthcoming initiations before formal disclosure.

However, this pattern could also reflect investors responding to observable financial signals, such as improving earnings or declining capital expenditure, that credibly predict an imminent initiation announcement without constituting private information. On balance, the evidence is more consistent with an anticipation effect based on public signals than with systematic information leakage, though this distinction cannot be definitively resolved using the current methodology.

For decreases, the gradual negative CAAR drift across the event windows is consistent with a slow information incorporation process, which may reflect the difficulty investors face in valuing the implications of a dividend cut in the absence of additional disclosures about the firm's financial condition.

5.6 Comparison with Prior Literature

The findings of this study are broadly consistent with the established empirical literature on dividend announcement effects, while also offering some notable points of comparison. Aharony and Swary (1980), in one of the earliest event studies of dividend announcements, document significant positive abnormal returns around unexpected dividend increase announcements and negative returns around decreases. The present study replicates the positive reaction to increases and the negative direction for decreases, though the decrease result does not achieve significance, which may reflect differences in sample composition and the longer and more recent study period used in this study.

The results for dividend initiations tell a slightly different from what earlier research found. Michaely, Thaler and Womack (1995) found that when firms start paying dividends for the first time, investors tend to react very positively.

This study did not find the same clear positive reaction. However, there is a good reason for this difference. The firms in this study are all S&P 500 firms and are large, well-known companies that analysts and investors follow very closely every day. For these types of companies, the decision to start paying a dividend is rarely a big surprise because investors already have access to a lot of information about the company's financial health.

For smaller or lesser-known companies, starting to pay a dividend would be much more unexpected and would likely generate a stronger reaction. This explains why the reaction to initiations in this study is more mixed and harder to interpret compared to what earlier studies found using broader samples of companies.

The near-zero regression coefficients and negligible R-squared values are consistent with the observations of Amihud and Li (2006), who document a decline in the informational content of dividend announcements over time, driven by the rise in institutional investor holdings. Because institutional investors are better informed and trade ahead of announcements, the surprise component that drives market reactions is reduced, particularly among heavily institutionally owned firms such as those in the S&P 500.

In a market like the S&P 500 where investors already have access to a huge amount of information about every company, a dividend announcement does not reveal as much new information as it would in a less transparent market. This is why firm characteristics like size and profitability cannot reliably predict how strongly the market will react to any individual announcement, by the time the announcement is made, most of what it could say has already been priced in. That said, the results for dividend increases still show a clear and consistent positive market reaction at the overall level, which confirms that dividend announcements still carry meaningful information even for these large and well-known companies. This supports the view that dividend signaling theory is still relevant today for large U.S. firms.

5.7 Chapter Summary

This chapter has discussed the empirical findings of the study in relation to the three research hypotheses, the cross-sectional regression analysis, and the broader implications for market efficiency and prior literature.

The evidence supports H1, with dividend increases generating a statistically significant positive cumulative abnormal return concentrated in the day immediately following the announcement. The absence of pre-announcement drift and the rapid stabilization of prices after the reaction confirm that the S&P 500 market processes publicly available dividend information quickly and completely, consistent with semi-strong form efficiency. The positive reaction also confirms that dividend increases carry genuine informational value in the modern market setting, signaling management confidence in the firm's earnings sustainability.

H2 is directionally consistent with prior evidence on dividend cut reactions but cannot be formally confirmed. The evidence suggests that the market does not treat all dividend cuts equally. Cuts driven by financial distress appear to generate negative reactions while those reflecting strategic capital reallocation are interpreted more neutrally or even positively. This heterogeneity in investor interpretation dilutes the average reaction and prevents a statistically reliable conclusion, even though the directional evidence remains meaningful.

H3 is not supported. Dividend initiations produce a complex intra-window pattern where pre-announcement price appreciation is followed by a sharp reversal on the announcement day itself. This pattern reflects the inherent economic ambiguity of an initiation signal, the transition from a growth firm to a dividend-paying firm carries different implications for different investors, and the mixed reaction observed here captures that tension directly.

The cross-sectional regression confirms that observable firm characteristics have no meaningful ability to predict the magnitude of individual announcement reactions. This finding points to a fundamental feature of how dividend signaling works in large, information-rich markets the reaction to any given announcement is shaped primarily by how surprising it is relative to investor expectations at that specific moment, not by how

large or profitable the firm is in general terms. Standard annual accounting data simply cannot capture that dimension.

Taken together, the results show that dividend announcements remain a relevant and informative corporate signal among S&P 500 firms, but their effect operates at the aggregate level rather than the individual event level. Chapter 6 discusses the limitations of the study and what they mean for how broadly these findings can be applied.

CHAPTER 6

LIMITATIONS

This chapter sets out the main limitations of the present study. Identifying these limitations is not an attempt to undermine the findings but to help the reader understand where they apply, where they may not, and what future research could do to address the gaps left here. The limitations are grouped into: sample scope and data constraints, methodology and statistical power.

6.1 Sample Scope and Data Constraints

The study is limited to firms that were constituents of the S&P 500 index as of 2nd February 2025. This means only large, well-established U.S. companies are included. Smaller firms, mid-cap companies, and firms that were once in the S&P 500 but later removed due to bankruptcy or declining size are not captured. This introduces what is known as survivorship bias.

Because the sample only includes companies that survived and remained large enough to be in the index throughout or by the end of the study period, the results may reflect a systematically more successful group of firms than the broader market. This could lead to an underestimation of negative market reactions, particularly for dividend decreases, if financially weaker firms that were delisted are missing from the analysis.

Further data constraint concerns the leverage variable. Debt-to-equity ratio was originally included in the cross-sectional regression as a control variable based on agency theory predictions. However, it had to be excluded from all regression models because the coverage in the dataset was insufficient too many firms were missing values for this variable, and including it would have reduced the usable sample to a size that was too small to produce reliable results. This means one theoretically important cross-sectional driver of dividend announcement reactions could not be tested. Future studies with more complete leverage data would be better placed to examine whether highly leveraged firms generate systematically different market reactions to dividend announcements.

Another data limitation concerns missing price observations for some firms in the sample. Out of 503 tickers examined, 91 firms had incomplete data over the research window from January 2011 to December 2025. The reasons for these gaps vary some firms had not yet been listed at the start of the study period due to recent IPOs, others were spun off from parent companies during the window, and some went through mergers, acquisitions or bankruptcy proceedings that interrupted their trading history.

Where the gaps were small enough, the event-by-event minimum observation threshold of 60 trading days ensured that only events with sufficient estimation data were included in the analysis. However, for firms with more extensive gaps, individual events were dropped from the sample entirely rather than excluded at the firm level, meaning the firm still contributed events where data was available. A list of the firms with incomplete data, including the specific reason for each gap, is provided in Appendix 1.

6.2 Methodology

The market model used in this study assumes a stable linear relationship between a firm's stock returns and the returns of the S&P 500 index over the estimation window. In practice, this relationship may not be perfectly stable as a firm's beta can shift over time due to changes in business strategy, capital structure, or market conditions. If the beta estimated during the estimation window does not accurately represent the firm's risk profile during the event window, the abnormal return estimates may be slightly over or understated.

6.3 Statistical Power

The most significant statistical limitation of the study is the small size of the dividend decrease and initiation subsamples. Only 244 decrease events and 379 initiation events were retained after filtering, compared to 4,614 increase events. With such small subsamples and high dispersion of individual event CARs particularly for decreases, where the standard deviation exceeds 8%, the hypothesis tests lack the statistical power to detect moderate-sized effects.

This is why H2 and H3 could not be confirmed statistically even though the results pointed in the theoretically expected directions for decreases and showed an interesting intra-window pattern for initiations. A larger sample of decrease and initiation events, drawn from a broader universe of firms or a longer study period, might provide more reliable evidence on these two hypotheses.

6.4 Absence of an Expectations Model

This study sorts announcements into increases, decreases, and initiations by comparing each dividend to the firm's previous one, but it does not estimate what the market expected the dividend to be. This matters because signaling theory says prices react to the part of an announcement that surprises investors, not to the change on its own (Bhattacharya, 1979; Miller and Rock, 1985). Many S&P 500 firms raise their dividends on a steady, predictable schedule (Lintner, 1956), so a large share of the increases in the sample were probably expected well before they were announced. The increase group therefore mixes surprising raises, which should move prices, with routine ones, which should not. This likely pulls the average reaction down, which means the true

reaction to genuinely unexpected increases is probably larger than the +0.27% reported in this study. Future work could address this by building a measure of the expected dividend, for instance from analyst forecasts or a Lintner-style adjustment model, to isolate the surprise component directly, as noted in Section 7.4.

Finally, the results of this study are specific to large-cap U.S. equities and should not be generalized without further evidence. Taken together, these limitations point toward clear directions for future research, which are discussed further in Chapter 7.

CHAPTER 7

CONCLUSION

This study set out to examine how the stock market reacts to dividend announcements made by S&P 500 firms and to explore whether firm-level characteristics can help explain why some announcements generate larger price reactions than others. Using a sample of 22,118 ordinary dividend announcements across 414 firms over the period from January 1st 2011 to December 31st 2025, the study applied a standard event study methodology based on the OLS market model to measure abnormal returns around each announcement.

Cross-sectional regression analysis was then used to test whether announcement type and firm characteristics such as size, profitability and valuation could explain variation in individual event returns. This chapter brings together the key findings, discusses what they contribute to the existing literature, and outlines what future research in this area could build on.

7.1 Summary of Key Findings

The clearest finding from this study is that dividend increases generate a genuine and statistically significant positive market reaction. The mean cumulative abnormal return for increases is +0.13% over the short (-1, +1) window, significant at the 10% level, and rises to +0.27% over the wider (-3, +3) window, where it is significant at the 1% level. The pattern of returns is also telling prices stay essentially flat in the days before the announcement and then rise sharply on the day after, which is consistent with how an efficient market should respond to new public information. This result supports H1 and is consistent with dividend signaling theory, which holds that dividend increases represent a credible signal of management's confidence in the firm's future earnings.

For dividend decreases, the results point in the expected negative direction, mean CARs of -0.41% and -0.60% over the two windows respectively, but neither result reaches statistical significance. The small size of the decrease subsample, with only 244 events and very high variation in individual returns, means the data does not have enough precision to formally confirm H2. The direction of the result is still meaningful and consistent with theory, but the evidence is not strong enough to draw a firm conclusion.

Dividend initiations produced the most interesting and complex pattern in the study. Rather than a simple positive reaction as H3 predicted, the CAAR for initiations builds steadily in the days before the announcement, reaches a peak on day -1, then drops sharply on the announcement day itself before gradually recovering. This pattern is consistent with a sell-the-news effect, where investors who bought shares in anticipation of the announcement sell once it is confirmed. Neither event window produces a statistically significant result, meaning H3 is not supported.

The cross-sectional regression found that none of the firm-level control variables, firm size, profitability, price-to-book ratio or dividend type produced statistically significant coefficients in any of the 3 model specifications.

The R-squared values remained below 0.2% across all models and both event windows. It tells us that while dividend announcements do move prices at the aggregate level, the size of the reaction in any individual announcement event is driven by factors that standard firm characteristics cannot explain such as how surprising the announcement was relative to market expectations, or whether other news was released at the same time.

7.2 Theoretical Contributions

This study makes several contributions to the academic understanding of dividend announcements and market efficiency. First, it provides updated evidence that dividend signaling theory remains relevant for large-cap U.S. firms in the post-2010 period. While some earlier research, particularly Amihud and Li (2006), suggested that the informational content of dividend announcements was declining over time, this study shows that increases still generate a consistent and statistically significant positive market reaction even in a period characterized by high market transparency and widespread analyst coverage.

Secondly, the results offer fresh evidence on market efficiency in the context of dividend announcements. The concentration of the price reaction on day +1 rather than day 0, the absence of pre-announcement drift for increases, and the immediate price stabilization after day +1 are all consistent with semi-strong form efficiency as described by Fama (1970). The market incorporates the new information from a dividend increase quickly and completely, with no systematic pattern left for investors to exploit after.

Third, the regression results show that when it comes to individual dividend announcements, the size of the market reaction cannot be predicted by looking at a firm's size, profitability or valuation. What seems to matter more is how surprising the announcement was. If the market was already expecting a dividend increase based on what it knew about the firm, the reaction will be small. If the announcement came as a genuine surprise, the reaction will be larger. Standard financial data captured in annual reports simply cannot tell you in advance how much of a surprise any given announcement will be, which is why the regression models explain almost nothing.

7.3 Practical Implications

The findings of this study has implications for investors, corporate managers and financial analysts. For investors, the results suggest that dividend increases among S&P 500 firms are associated with a small but genuine positive market reaction that is concentrated in a very short window around the announcement. However, the finding that firm characteristics such as size and profitability cannot predict the size of this reaction means that no simple strategy of buying shares in firms with certain financial profiles ahead of their dividend announcements is likely to generate consistent returns. The reaction is too driven by event-specific surprise to be systematically exploited using publicly available firm-level data.

For corporate managers and boards, the results reinforce the view that dividend decisions are closely watched by the market and carry informational weight. A dividend increase is likely to be received positively, particularly if it is not already anticipated. The results for initiations suggest that companies considering their first dividend payment should be mindful of investor expectations, if the market has already priced in a positive announcement based on observable financial improvements, the announcement itself may not deliver the price boost management might expect.

For financial analysts, the regression results highlight the limitations of using standard annual accounting metrics to forecast the market impact of individual dividend announcements. The surprise component of an announcement, how it compares to what investors were expecting appears to matter far more than the firm's size, profitability or valuation at the time. Incorporating analyst forecast data and measuring the degree of announcement surprise would likely produce more informative models of cross-sectional variation in dividend announcement returns.

7.4 Recommendations

From the findings and limitations of this study, the most immediate priority is to address the small sample problem for dividend decreases and initiations. A broader sample drawn from a wider universe of U.S. firms, or a longer historical period, would provide more statistical power to test H2 and H3 with greater confidence. The decrease subsample in particular would benefit from a larger dataset given the high heterogeneity of market reactions observed within this group.

A second important extension would be to incorporate analyst forecast data to construct a direct measure of announcement surprise. The near-zero regression results in this study suggest that the degree to which an announcement exceeds or falls short of market expectations is a more important driver of the market reaction than the firm's observable financial characteristics. Studies that measure surprise directly using analyst consensus forecasts are likely to explain a much larger share of the cross-sectional variation in event returns.

The findings of this study are specific to large-cap U.S. firms in the S&P 500 over the period 2011 to 2025. These are among the most heavily covered and closely watched companies in the world, operating in one of the most information-efficient equity markets. This means the market reactions observed here may partly reflect a market that has already partially priced in the announcement before it is formally made. Whether the same

patterns hold for smaller, less-covered companies, firms in emerging markets, or firms operating in less liquid markets is an open question. A natural extension of this research would be to replicate the analysis using a broader cross-section of firms across different size categories and geographies, which would allow a direct test of whether dividend signaling effects are stronger where information asymmetry between managers and investors is higher.

Finally, future research could examine whether the findings differ across economic cycles. The study period from 2011 to 2025 includes periods of sustained economic growth, the COVID-19 shock of 2020, and rising interest rates from 2022 onwards. Testing whether dividend announcement effects are stronger or weaker in different macroeconomic environments would add important context to the findings presented here.

7.5 Closing Statement

This study set out to answer a straightforward question, do dividend announcements move stock prices, and if so, does it depend on what kind of announcement it is and what kind of firm is making it? The answer to the first part is yes, at least for dividend increases. The answer to the second part is more complex, the type of announcement matters at the aggregate level, but observable firm characteristics cannot explain why some events generate larger reactions than others. These findings confirm that dividend increase announcements remain a meaningful source of market information among S&P 500 firms in the modern era, consistent with the predictions of dividend signaling theory and with the principles of semi-strong form market efficiency. They also point clearly toward where the next generation of research in this area should focus on surprise, context and the conditions under which dividend signals carry the most weight.

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Appendix

Appendix 1: S&P 500 Firms with Incomplete Price Data (N = 91)

No.	Ticker	Missing Period	Missing Obs.	Reason for Missing Data
1	ABBV-US	Jan 2011 to Dec 2012	487	Spun off from Abbott Laboratories in January 2013
2	ABNB-US	Jan 2011 to Dec 2020	2,502	IPO December 2020
3	ALLE-US	Jan 2011 to Nov 2013	724	Spun off from Ingersoll-Rand in December 2013
4	GOOG-US	Jan 2011 to Apr 2014	817	Class C shares created in April 2014 stock split
5	AMCR-US	Jan 2011 to Jun 2019	2,122	Listed on NYSE June 2019 after merger with Bemis
6	APO-US	Jan 2011 to Mar 2011	60	Converted from limited partnership to corporation; data gap reflects restructuring
7	APP-US	Jan 2011 to Apr 2021	2,587	IPO April 2021
8	APTV-US	Jan 2011 to Nov 2011	222	Bankrupt 2005 /Created when Delphi Automotive split into two companies in 2017
9	ARES-US	Jan 2011 to May 2014	837	IPO May 2014
10	ANET-US	Jan 2011 to Jun 2014	861	IPO June 2014
11	XYZ-US	Jan 2011 to Nov 2015; Dec 2018 to Dec 2025	3,004	Formerly Square Inc.; IPO November 2015
12	CB-US	Sep 2020 to Dec 2025	1,319	A segment of price data from extraction
13	CDW-US	Jan 2011 to Jun 2013	624	IPO July 2013
14	COO-US	Oct 2025 to Dec 2025	45	A segment of price data from extraction
15	CRH-US	Sep 2020 to Dec 2025	1,319	Requires further research
16	CARR-US	Jan 2011 to Mar 2020	2,317	Spun off from United Technologies in April 2020
17	CVNA-US	Jan 2011 to Apr 2017	1,590	IPO April 2017
18	CFG-US	Jan 2011 to Sep 2014	937	IPO September 2014
19	COIN-US	Jan 2011 to Apr 2021	2,586	Direct listing on Nasdaq April 2021
20	CEG-US	Jan 2011 to Jan 2022	2,780	Spun off from Exelon Corporation February 2022
21	CTVA-US	Jan 2011 to May 2019	2,111	Spun off from DowDuPont June 2019
22	DD-US	Sep 2020 to Dec 2025	1,319	A segment of price data from extraction
23	DDOG-US	Jan 2011 to Sep 2019	2,192	IPO September 2019
24	DELL-US	Jan 2011 to Aug 2016	1,410	Dell went private 2013; relisted December 2018
25	FANG-US	Jan 2011 to Oct 2012	449	IPO October 2012
26	DASH-US	Jan 2011 to Dec 2020	2,501	IPO December 2020
27	DOW-US	Jan 2011 to Mar 2019	2,065	Spun off from DowDuPont April 2019

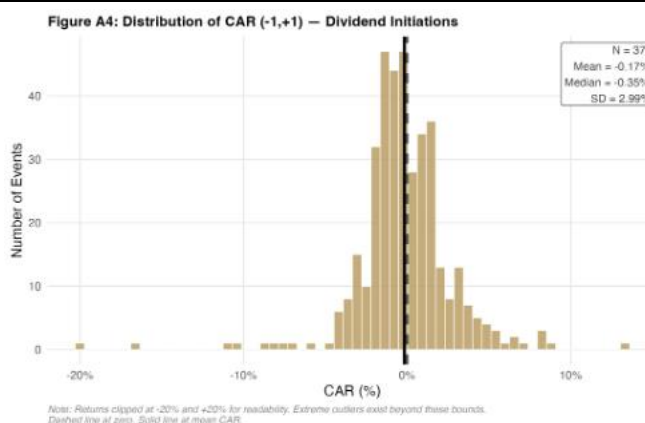
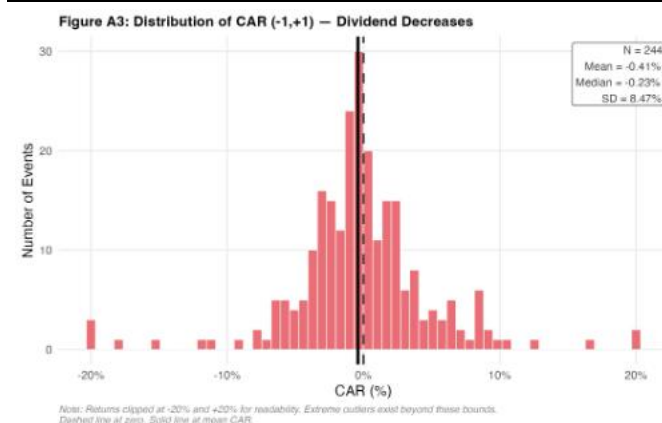
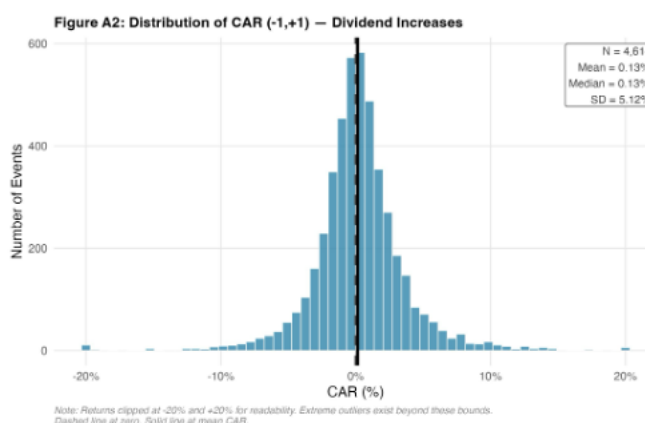
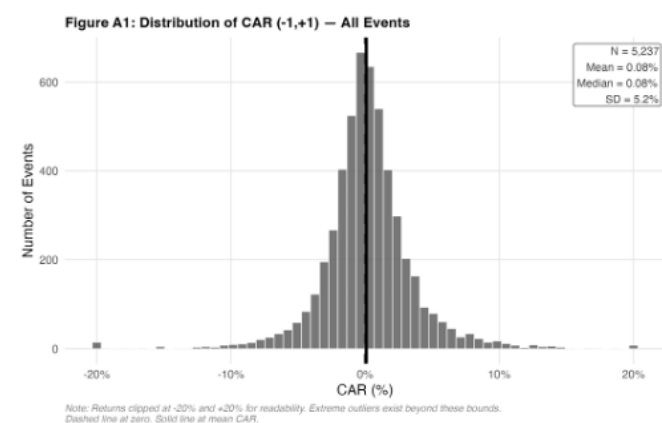
28	EPAM-US	Jan 2011 to Feb 2012	277	IPO February 2012
29	EXPE-US	Sep 2020 to Dec 2025	1,319	Requires further research
30	EXE-US	Jan 2011 to Feb 2021	2,543	Formerly Chesapeake Energy; emerged from bankruptcy 2021, renamed 2024
31	FTV-US	Jan 2011 to Jun 2016	1,369	Spun off from Danaher Corporation July 2016
32	FOXA-US	Jan 2011 to Feb 2019	2,051	Spun off when Disney acquired 21st Century Fox; March 2019
33	FOX-US	Jan 2011 to Feb 2019	2,051	Spun off when Disney acquired 21st Century Fox; March 2019
34	GEHC-US	Jan 2011 to Dec 2022	3,009	Spun off from General Electric January 2023
35	GEV-US	Jan 2011 to Mar 2024	3,329	Spun off from General Electric April 2024
36	GDDY-US	Jan 2011 to Apr 2015	1,067	IPO April 2015
37	HCA-US	Jan 2011 to Mar 2011	46	Re-IPO March 2011 after going private in 2006
38	HPQ-US	Sep 2020 to Dec 2025	1,319	HP Inc.; spun off from Hewlett-Packard November 2015
39	HPE-US	Jan 2011 to Oct 2015	1,206	Spun off from Hewlett-Packard Company November 2015
40	HLT-US	Jan 2011 to Dec 2013	741	IPO December 2013
41	HWM-US	Jan 2011 to Oct 2016	1,458	Formerly Arconic; spun off from Alcoa 2016 and restructured
42	HII-US	Jan 2011 to Mar 2011	54	Spun off from Northrop Grumman March 2011
43	IQV-US	Jan 2011 to May 2013	590	Formed from merger of IMS Health and Quintiles Transnational in 2016
44	IR-US	Jan 2011 to May 2017	1,600	Formed from Gardner Denver and Ingersoll-Rand merger February 2020
45	INVH-US	Jan 2011 to Feb 2017	1,530	IPO February 2017
46	KVUE-US	Jan 2011 to May 2023	3,104	Spun off from Johnson & Johnson May 2023
47	KEYS-US	Jan 2011 to Oct 2014	955	Spun off from Agilent Technologies November 2014
48	KMI-US	Jan 2011 to Feb 2011	28	Re-listed on NYSE February 2011; missing first weeks of sample
49	KHC-US	Jan 2011 to Sep 2012	430	Formed from merger of Kraft Foods and H.J. Heinz July 2015
50	LIN-US	Sep 2020 to Dec 2025	1,319	Linde plc; formed from merger of Linde AG and Praxair October 2018
51	LITE-US	Jan 2011 to Jul 2015	1,145	Spun off from JDSU (Viavi Solutions) August 2015
52	MOS-US	Sep 2020 to Dec 2025	1,319	A segment of price data from extraction
53	MSI-US	Sep 2020 to Dec 2025	1,319	A segment of price data from extraction
54	MPC-US	Jan 2011 to Jun 2011; Dec 2018 to Dec 2025	1,894	A segment of price data from extraction
55	META-US	Jan 2011 to May 2012; Dec 2018 to Dec 2025	2,122	A segment of price data from extraction
56	MRNA-US	Jan 2011 to Dec 2018; Dec 2018 to Dec 2025	3,771	IPO December 2018
58	NXPI-US	Dec 2018 to Dec 2025	1,775	A segment of price data from extraction

59	NWSA-US	Jan 2011 to Jun 2013; Dec 2018 to Dec 2025	2,393	New News Corp created when 21st Century Fox separated June 2013
61	OTIS-US	All data missing within research window	3,772	Spun off from United Technologies in April 2020
62	PNR-US	Sep 2020 to Dec 2025	1,319	A segment of price data from extraction
63	PPL-US	Dec 2018 to Dec 2025	1,775	A segment of price data from extraction
64	PLTR-US	All data missing within research window	3,772	Direct listing on NYSE September 2020
65	PANW-US	Jan 2011 to Jul 2012; Dec 2018 to Dec 2025	2,165	IPO July 2012
66	PSKY-US	Dec 2018 to Dec 2025	1,775	A segment of price data from extraction
67	PYPL-US	Jan 2011 to Jul 2015; Dec 2018 to Dec 2025	2,907	Spun off from eBay July 2015
68	PSX-US	Jan 2011 to Apr 2012; Dec 2018 to Dec 2025	2,096	Spun off from ConocoPhillips May 2012
69	Q-US	All data missing within research window	3,772	IPO 2021
70	HOOD-US	Jan 2011 to Jul 2021	2,660	IPO July 2021
71	STE-US	Sep 2020 to Dec 2025	1,319	A segment of price data from extraction
72	SW-US	Sep 2020 to Dec 2025	1,319	A segment of price data from extraction
73	SNDK-US	All data missing within research window	3,772	SanDisk relisted February 2025 after spin-off from Western Digital
74	SOLV-US	All data missing within research window	3,772	Spun off from 3M April 2024
75	SYF-US	Jan 2011 to Jul 2014; Dec 2018 to Dec 2025	2,674	Spun off from GE Capital (General Electric) July 2014
76	TEL-US	Sep 2020 to Dec 2025	1,319	A segment of price data from extraction
77	TKO-US	Oct 2025 to Dec 2025	45	A segment of price data from extraction
78	TRGP-US	Dec 2018 to Dec 2025	1,775	A segment of price data from extraction
79	TSLA-US	Dec 2018 to Dec 2025	1,775	A segment of price data from extraction
80	TTD-US	Jan 2011 to Sep 2016; Dec 2018 to Dec 2025	3,214	IPO September 2016
81	UBER-US	All data missing within research window	3,772	IPO May 2019
82	VICI-US	Jan 2011 to Oct 2017; Dec 2018 to Dec 2025	3,485	IPO February 2018; REIT spun out of Caesars Entertainment
83	VTRS-US	Sep 2020 to Dec 2025	1,319	Requires further research
84	VLTO-US	All data missing within research window	3,772	Spun off from Danaher Corporation September 2023
85	VRT-US	Jan 2011 to Jul 2018; Dec 2018 to Dec 2025	3,680	Listed via SPAC merger February 2020
86	VST-US	Jan 2011 to Sep 2016; Dec 2018 to Dec 2025	3,221	Emerged from Energy Future Holdings bankruptcy October 2016
87	WBD-US	Oct 2025 to Dec 2025	45	A segment of price data from extraction
88	WDAY-US	Jan 2011 to Oct 2012; Dec 2018 to Dec 2025	2,224	IPO October 2012
89	XYL-US	Jan 2011 to Oct 2011; Dec 2018 to Dec 2025	1,972	Spun off from ITT Corporation October 2011

90	ZBH-US	Dec 2018 to Dec 2025	1,775	A segment of price data from extraction
91	ZTS-US	Jan 2011 to Feb 2013; Dec 2018 to Dec 2025	2,298	Spun off from Pfizer February 2013

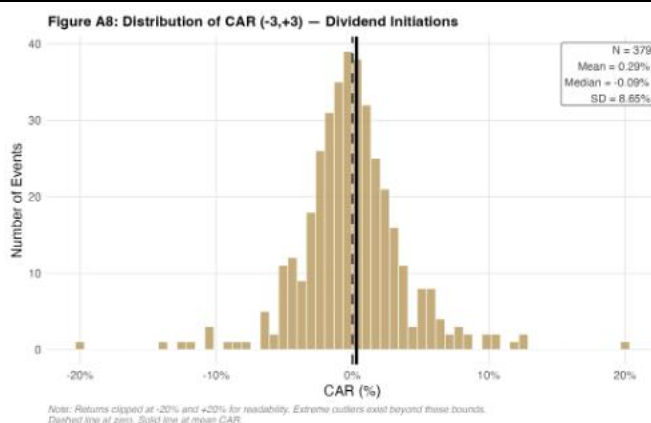
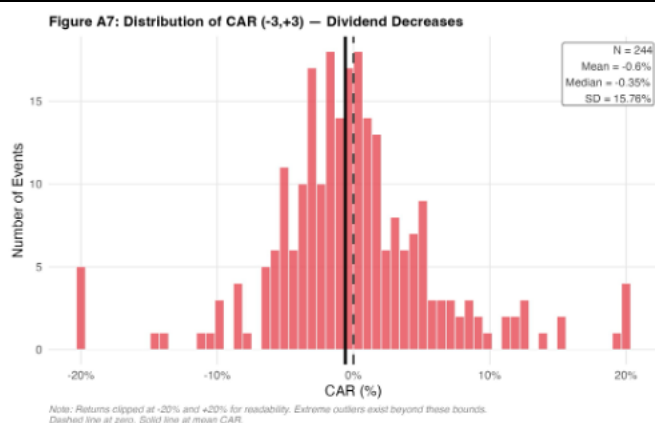
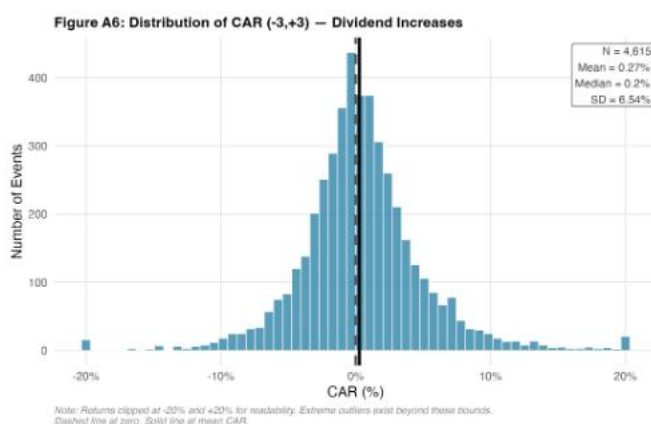
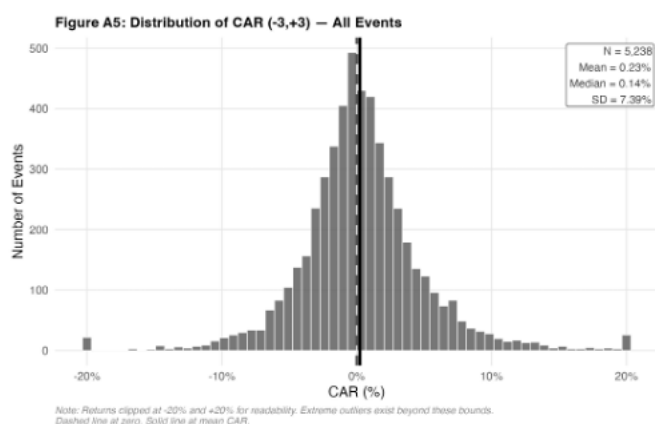
Note: Missing Obs. = number of trading days with no valid price data within the research window (January 2011 to December 2025). Firms classified as Incomplete had insufficient data for at least one portion of the study period. Where the reason is listed as Requires further research, the exact cause of the data gap could not be confirmed from available sources. These firms still contributed events to the analysis where sufficient estimation window data was available.

Appendix 2A: CAR Distribution Histograms Event Window (-1, +1)



Note: Figures A1 to A4 show the distribution of individual event CARs over the (-1, +1) event window for all events and each announcement subsample. Returns are clipped at -20% and +20% for readability. Extreme outliers exist beyond these bounds. Dashed vertical line at zero. Solid vertical line at mean CAR. All values in percentages.

Appendix 2B: CAR Distribution Histograms Event Window (-3, +3)



Note: Figures A5 to A8 show the distribution of individual event CARs over the (-3, +3) event window for all events and each announcement subsample. Returns are clipped at -20% and +20% for readability. Extreme outliers exist beyond these bounds. Dashed vertical line at zero. Solid vertical line at mean CAR. All values in percentages.